



PREDICTING THE COST OF YOUR INTERNATIONAL EDUCATION JOURNEY



Your Future at University
Cost Analysis & Machine Learning
Predictions





Meet the Team – Group 5

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PROBLEM ?



“A massive IDP survey of 21,000 students found that **only 55% of students had general knowledge of the actual money they would need to go abroad for studies**”



A global survey from
IDP Education

Source

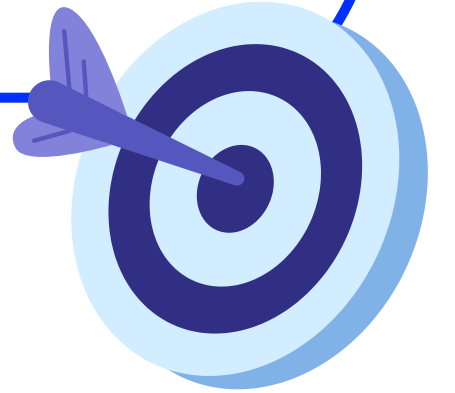


What are the Objectives ??



- Our Main Objective -

Build a predictive model that can **estimate the total cost of your international education based on your specific choices**, your degree level, your field of study, your target university, the city's living expenses, currency exchange realities, and program duration



But before we can predict, we must understand. And understanding begins with exploration.

Specifically, the EDA aims to

1. Assess cost variations across geographies and institutions
2. Compare costs across academic categories
3. Identify major cost drivers
4. Visualize trends and anomalies

The End **GOAL**

Transforming Your Choices Into Accurate Cost Predictions

Your Information

Advanced ML Model

Total Cost Prediction

ABOUT THE DATASET

- This Cost of International Education dataset is from Kaggle
- It compiles detailed financial information for students pursuing higher education abroad.
- It also covers multiple countries, cities, and universities around the world.

The dataset consists of **907 observations** and
12 variables

12 Variables

Qualitative

5

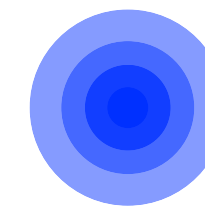
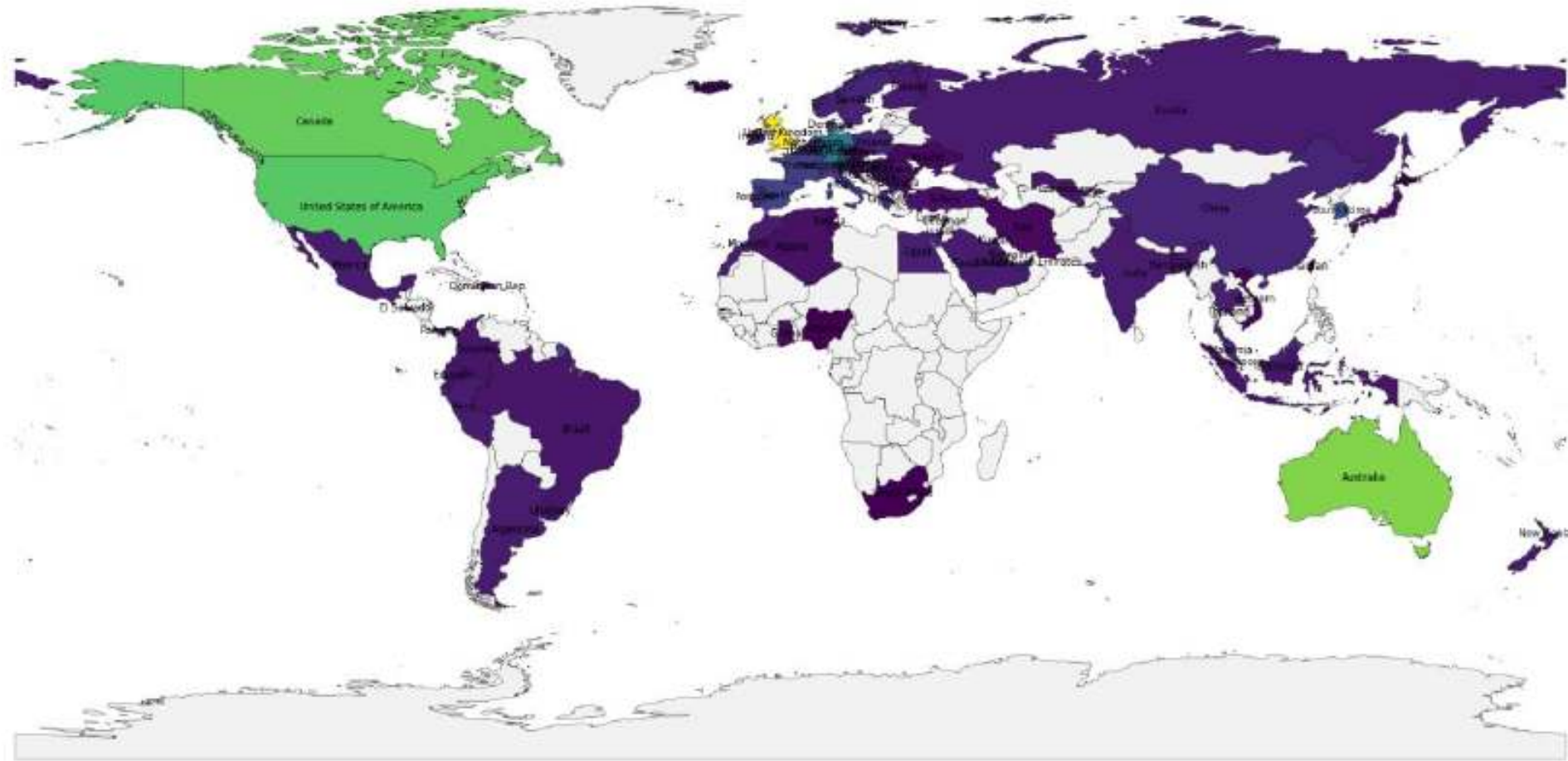
Country
City
University
Program
Level

Quantitative

7

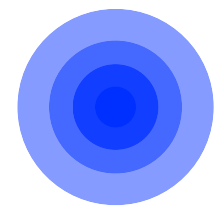
Duration_Years
Living_Cost_Index
Tuition_USD
Rent_USD
Visa_Fee_USD
Insurance_USD
Exchange_Rate

Global Distribution of Students



COUNTRY

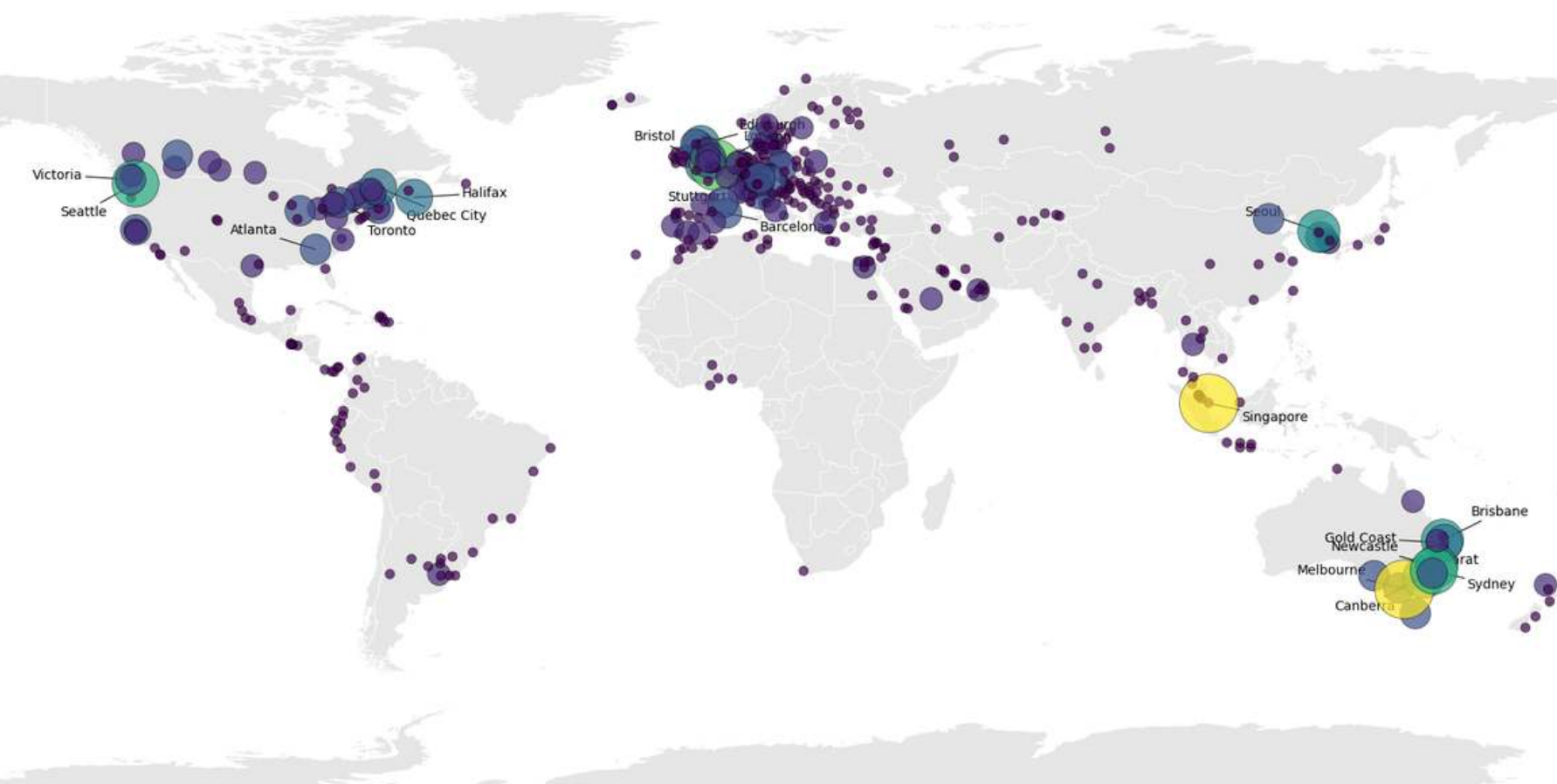
- ISO country name where the university is located.
- The dataset includes universities from **71 different countries, providing broad global coverage.**

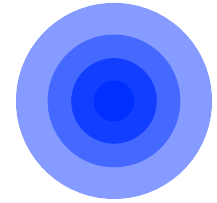


CITY

- City in which the institution sits.
- This variable represents **556 unique cities** in the dataset.

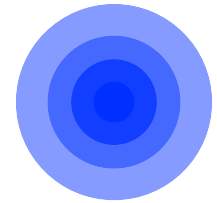
Global Distribution of Students — City Locations





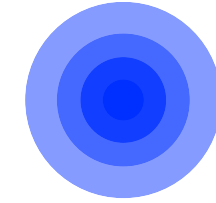
UNIVERSITY

- Official name of the higher-education institution (e.g., “Technical University of Munich”).
- The dataset includes **622 unique universities**



LEVEL

- Degree level of the program: **“Undergraduate”, “Master’s” or “PhD”.**



PROGRAM

- Specific course or academic major offered by the university (e.g., Master of Computer Science, MBA).
- The dataset includes **92 distinct programs**, covering a wide range of specializations.

Categorical Variable Summary

	count	unique	top	freq
University	907	622	University of Washington	6
Program	907	92	Computer Science	312
Level	907	3	Master	451

DURATION_YEARS

- Length of the program in years (e.g., 2 for a typical Master's).

RENT_USD



- Average **monthly** student accommodation rent in U.S. dollars.

VISA_FEE_USD



- **One-time visa application fee** paid by international students, in U.S. dollars.

TUITION_USD



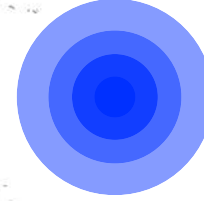
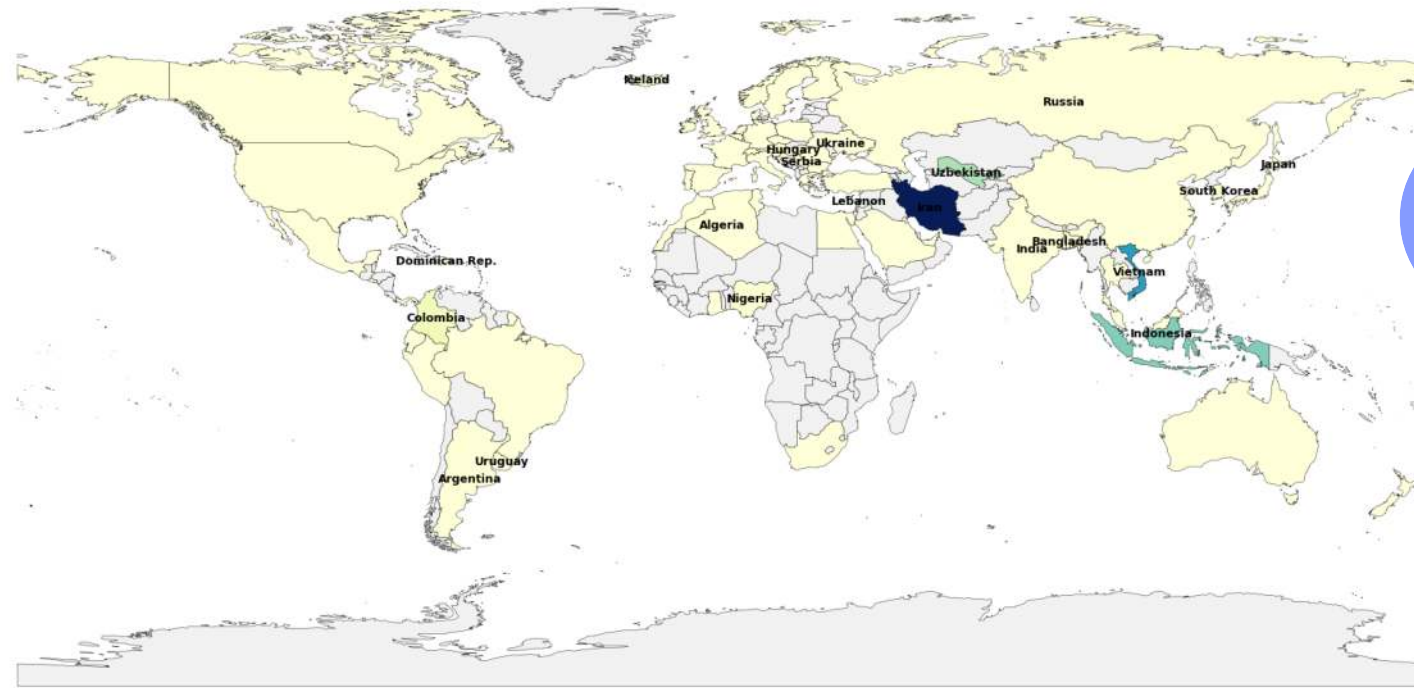
- **Total program tuition cost**, converted into U.S. dollars for ease of comparison

INSURANCE_USD



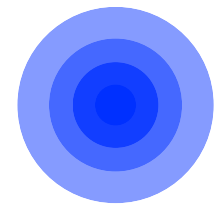
- **Annual** health or student insurance cost in U.S. dollars.

Global Exchange Rate by Country



EXCHANGE RATE

- **Local currency units per U.S. dollar** at the time of data collection vital for currency conversion and trend analysis if rates fluctuate.

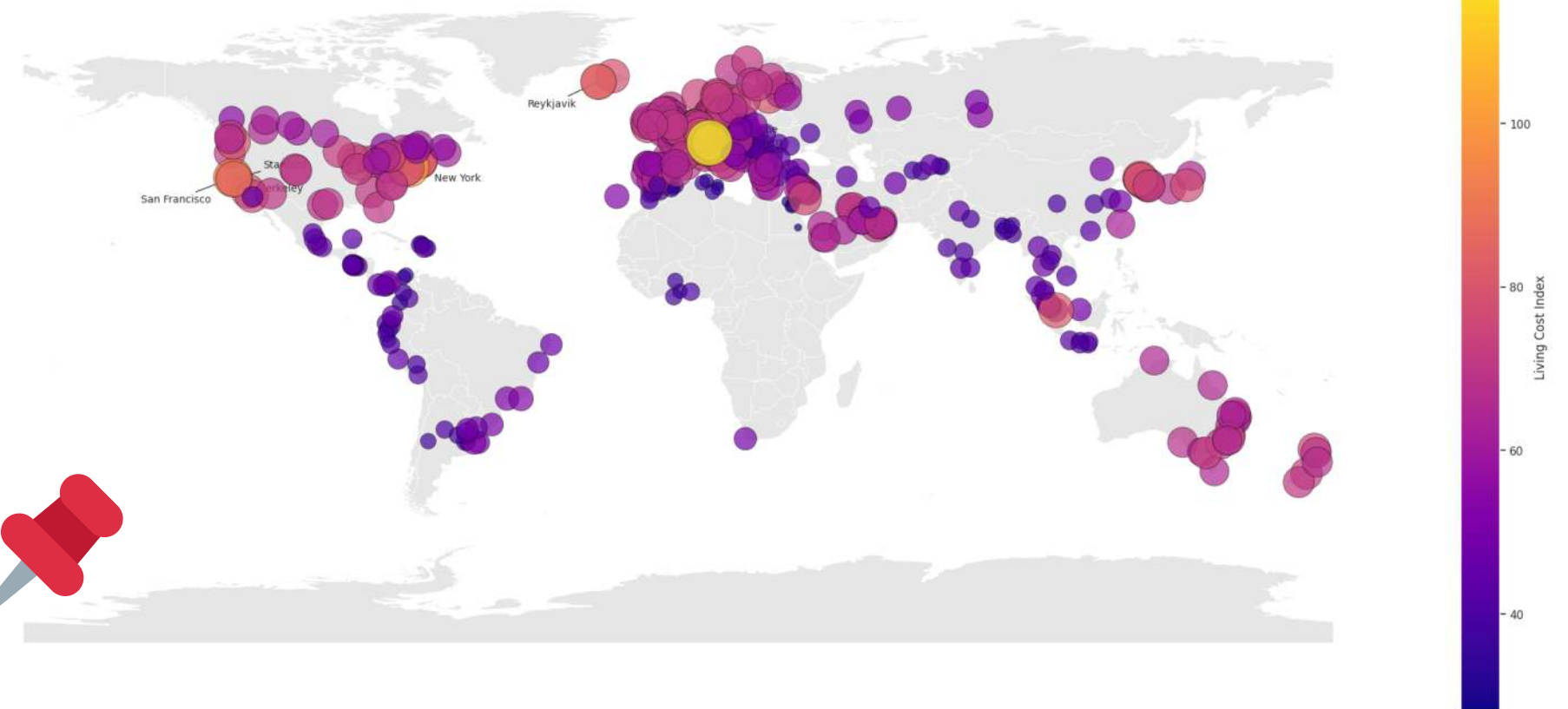


LIVING COST INDEX

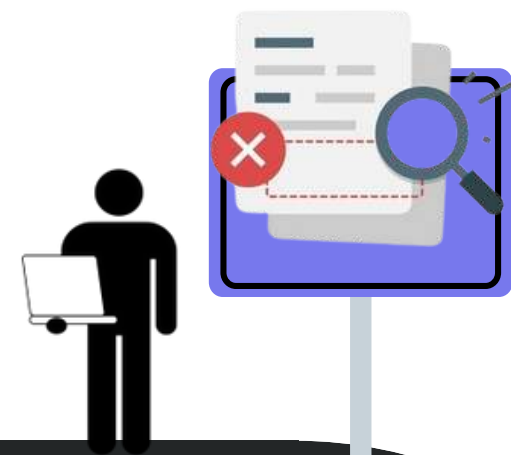
- **Normalized index reflecting relative day-to-day living expenses** (food, transport, utilities), with **New York City as the baseline (100)**.
Sourced from global cost-of-living databases.



Global Distribution of Cities — Living Cost Index



DATA CLEANING



-Data Types-

```
EDA
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 634 entries, 0 to 633
Data columns (total 13 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Country              634 non-null   object
1   City                 634 non-null   object
2   University           634 non-null   object
3   Program              634 non-null   object
4   Level                634 non-null   object
5   Duration_Years       634 non-null   float64
6   Tuition_USD          634 non-null   int64
7   Living_Cost_Index    634 non-null   float64
8   Rent_USD             634 non-null   int64
9   Visa_Fee_USD        634 non-null   int64
10  Insurance_USD        634 non-null   int64
11  Total_Cost_USD       634 non-null   float64
12  Exchange_Rate        634 non-null   float64
dtypes: float64(4), int64(4), object(5)
memory usage: 64.5+ KB
```

01

Check for Missing Values

No Missing Values

02

Check for Duplicates

No Duplicates

03

Check and Change Data Types

No Change

04

Check Unrealistic Values

Not found

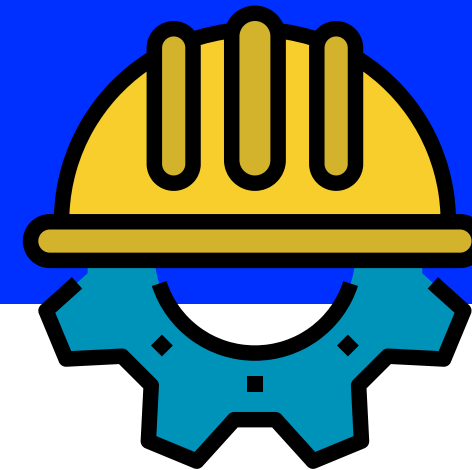
-Summary statistics of variables-

	Duration_Years	Tuition_USD	Living_Cost_Index	Rent_USD	Visa_Fee_USD	Insurance_USD	Exchange_Rate	Total_Cost_USD
count	907.000000	907.000000	907.000000	907.000000	907.000000	907.000000	907.000000	907.000000
mean	2.836825	16705.016538	64.437486	969.206174	211.396913	700.077178	623.000695	52218.111356
std	0.945449	16582.385275	14.056333	517.154752	143.435740	320.374875	3801.746134	37459.163755
min	1.000000	0.000000	27.800000	150.000000	40.000000	200.000000	0.150000	5700.000000
25%	2.000000	2850.000000	56.300000	545.000000	100.000000	450.000000	0.920000	22644.500000
50%	3.000000	7500.000000	67.500000	900.000000	160.000000	650.000000	1.350000	40020.000000
75%	4.000000	31100.000000	72.200000	1300.000000	240.000000	800.000000	7.150000	75600.000000
max	5.000000	62000.000000	122.400000	2500.000000	490.000000	1500.000000	42150.000000	199860.000000

	Country	City	University	Program	Level
count	907	907	907	907	907
unique	71	556	622	92	3
top	UK	Singapore	University of Washington	Computer Science	Master
freq	93	18	6	312	451



Feature Engineering



CREATE TRAGET VARIABLE

- The target variable is **Total_Cost_USD** representing the full cost for a student who is going to study abroad.



Tuition USD



Rent USD $\times$ 12 $\times$ Duration



(Living Cost Index / 100) $\times$ 12000
 $\times$ Duration



Visa Fee USD



Insurance USD $\times$ Duration



Total Cost USD

$$\begin{aligned} \text{Total_Cost_USD} = & \text{Tuition_USD} \\ & + (\text{Living_Cost_Index}/100 \times 12000 \times \text{Duration_Years}) \\ & + (\text{Rent_USD} \times 12 \times \text{Duration_Years}) \\ & + \text{Visa_Fee_USD} + (\text{Insurance_USD} \times \text{Duration_Years}) \end{aligned}$$



$$(\text{Living Cost Index} / 100) \times 12000 \times \text{Duration}$$

100

\$12000

- The Cost of Living Index (COLI) is often expressed relative to a baseline city, here it is **New York City = 100.**

- Baseline annual living cost of **~USD 12,000** estimated by summing **essential non-accommodation expenses (taken from a resource)**, using New York as the reference point.

$$\begin{aligned} \text{Monthly Total} &= \text{Transportation} + \text{Food} + \text{Utilities} + \text{Groceries} + \text{Clothes} \\ &= 128 + 638 + 116 + 116 + 42 \\ &= 1,040 \text{ USD/month} \end{aligned}$$

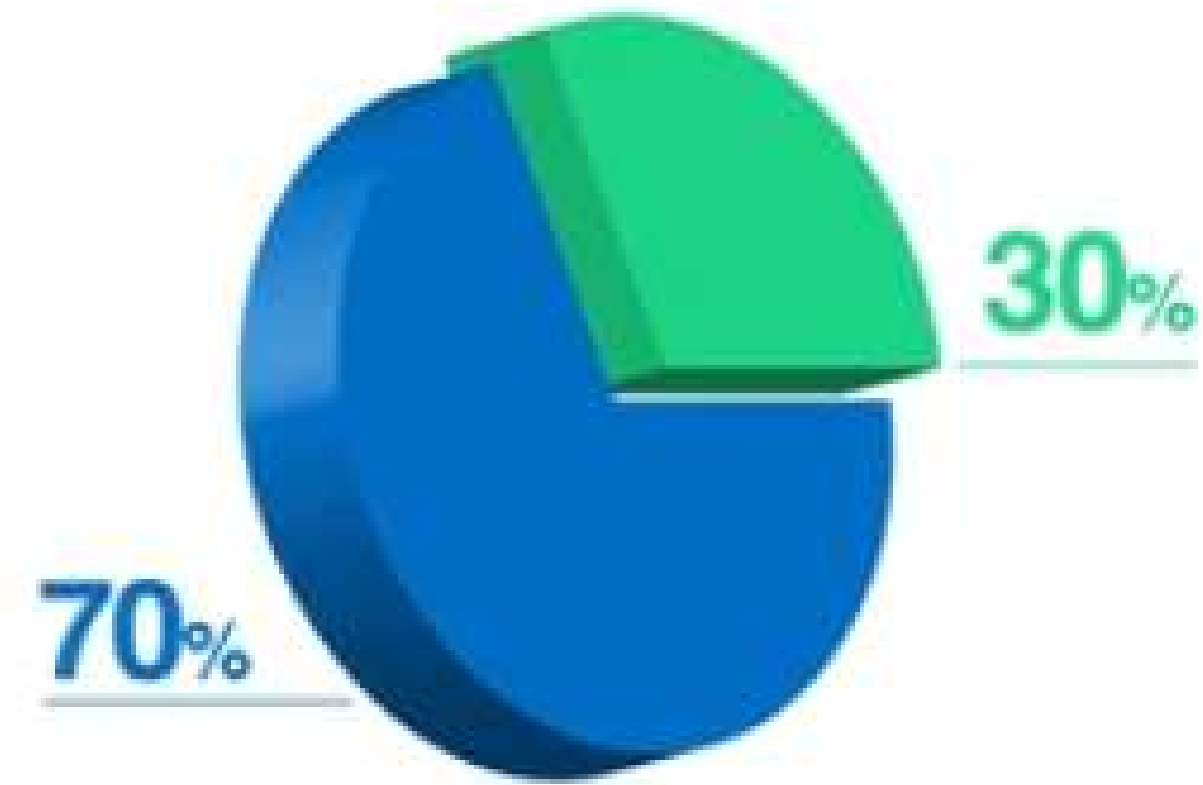
$$\begin{aligned} \text{Annual Total} &= \text{Monthly Total} \times 12 \\ &= 1,040 \times 12 \\ &= 12,480 \text{ USD/year} \end{aligned}$$

Rounded Approximation $\approx$ 12,000 USD/year

Particulars	Cost (Monthly)
Off-Campus Accommodation	\$1,050
On-Campus Accommodation	\$1,252
Transportation	\$128
Food Cost	\$638
Utility Cost	\$116
Entertainment	\$65
Groceries	\$116
Clothes	\$42
Health Insurance	\$80
Total Cost of Living with Off-Campus Accommodation	\$2,027
Total Cost of Living with On-Campus Accommodation	\$2,229
Annual Cost of Living	\$24,324

• **Living_Cost_Index**
Normalized index reflecting relative day-to- day living expenses (food, transport, utilities), with New York City as the baseline (100).
Sourced from global cost-of-living databases.

Train–Test Split of the Dataset



Splitting Strategy Used :

- 70% Training data
- 30% Testing data
- Random split with a fixed random seed to ensure reproducibility (random_state =1234)

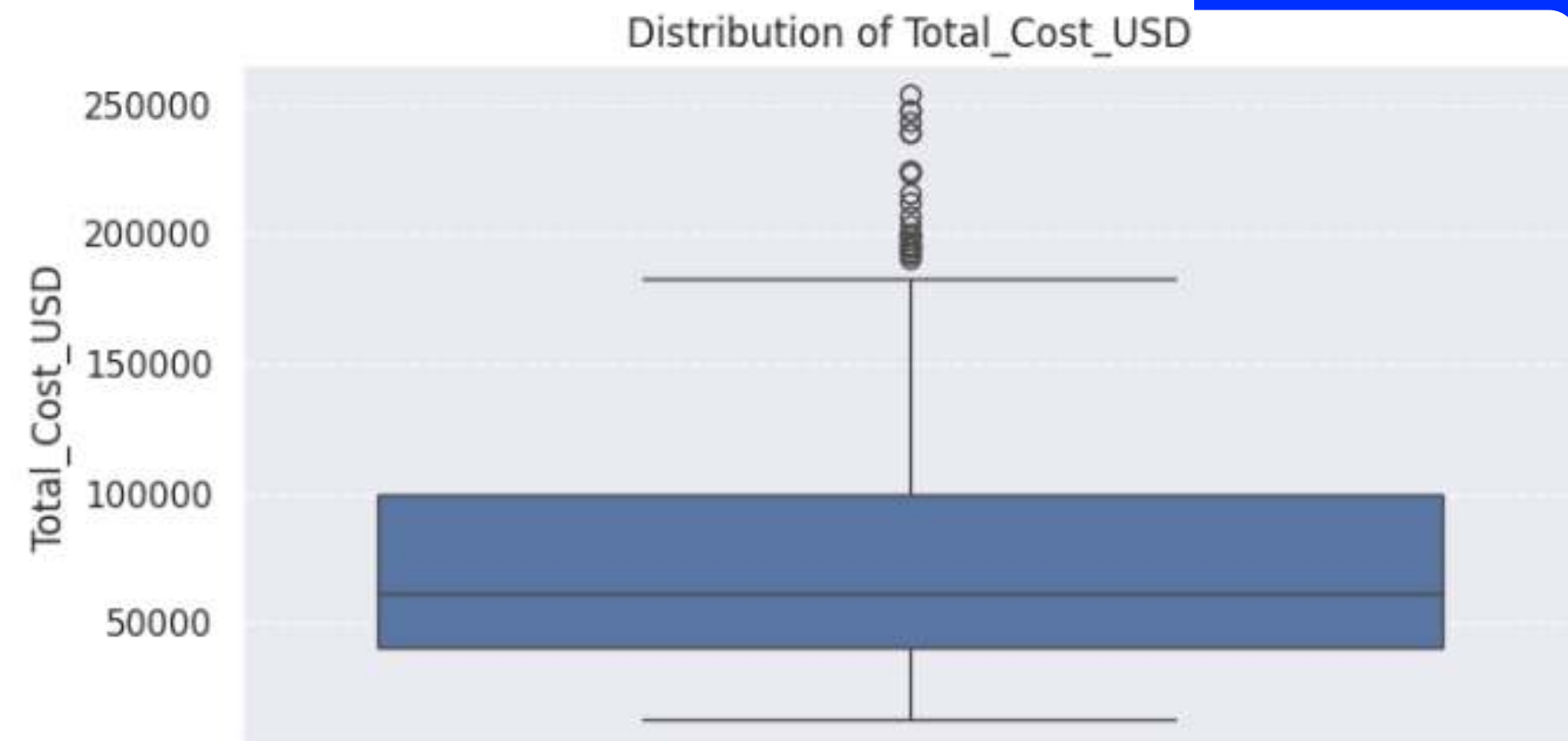
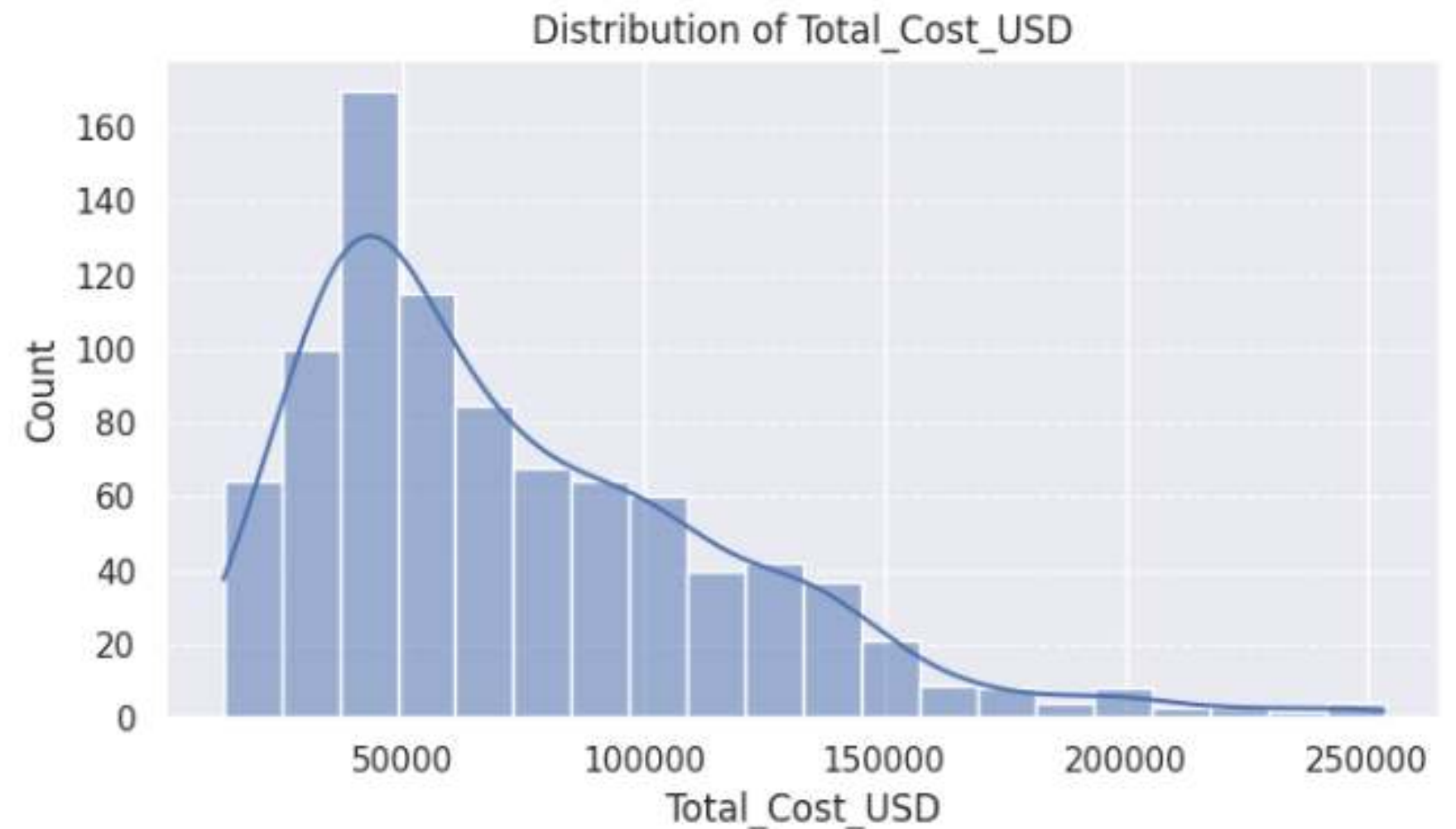
The training dataset has 634 rows and 13 columns.
The testing dataset has 273 rows and 13 columns.

Let's Look into EDA!

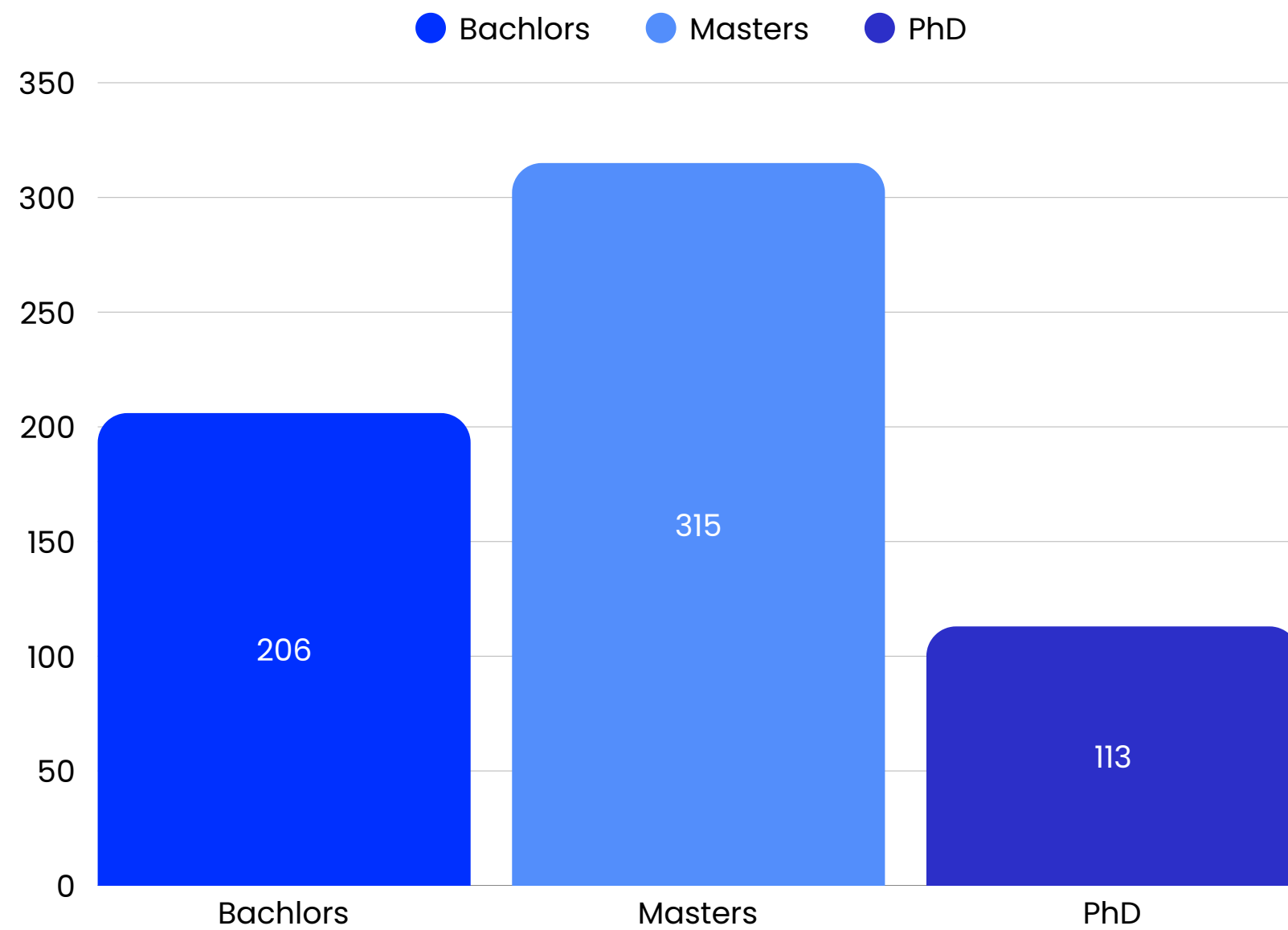


DISTRIBUTION OF TOTAL COST (USD)

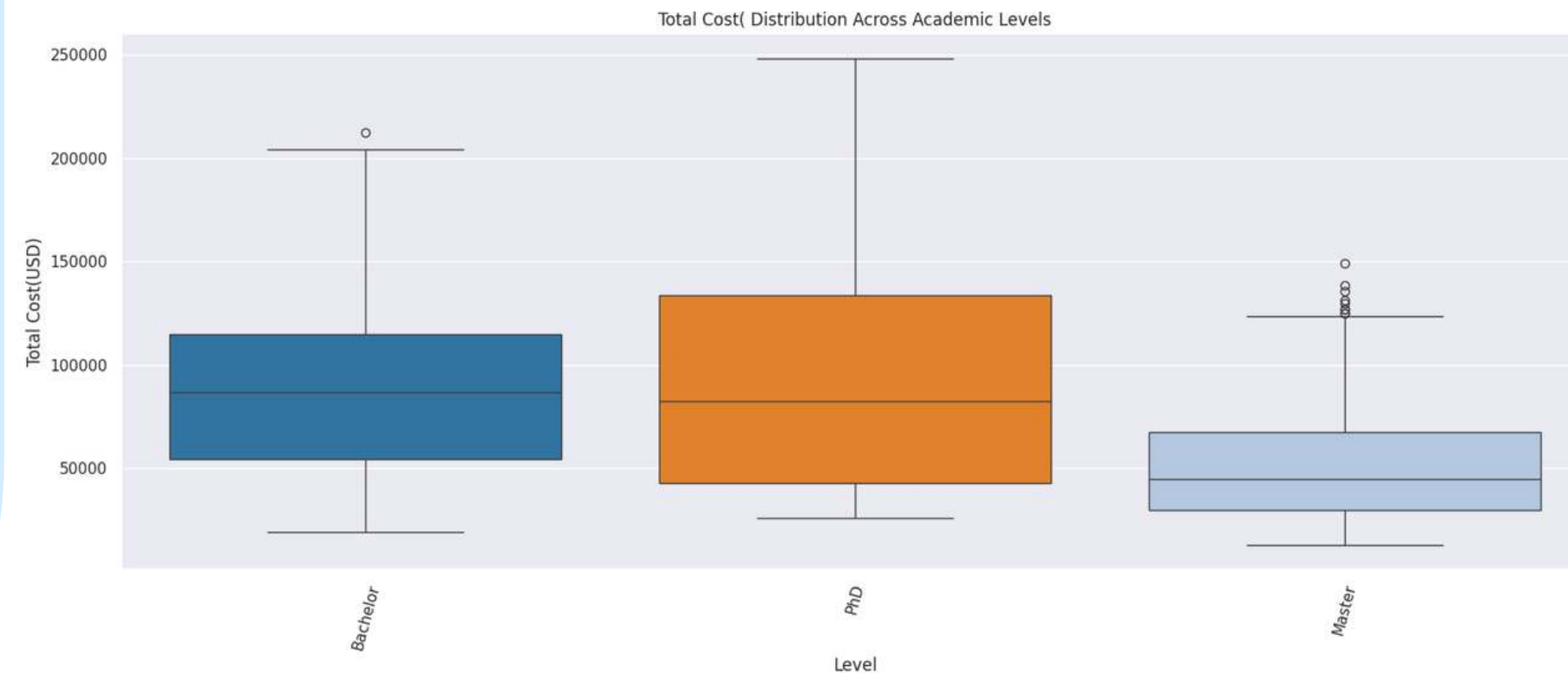
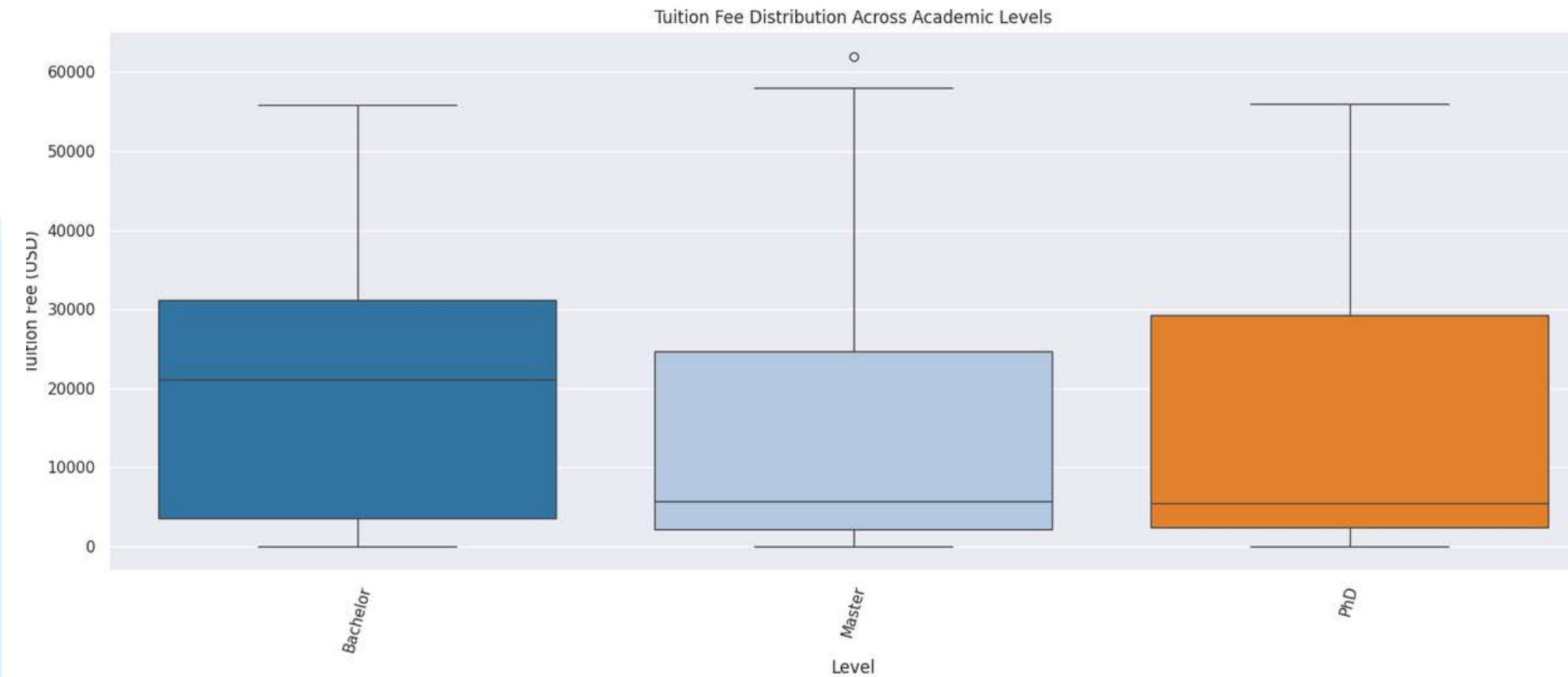
- The data shows a **strong right-skewed (1.26) distribution**, indicating most costs are concentrated at lower values with a long tail extending toward higher costs
- The majority of cases fall between approximately **\$25,000-\$150,000**
- About **75% of costs fall below \$100,000** (based on the upper quartile)



• LEVEL

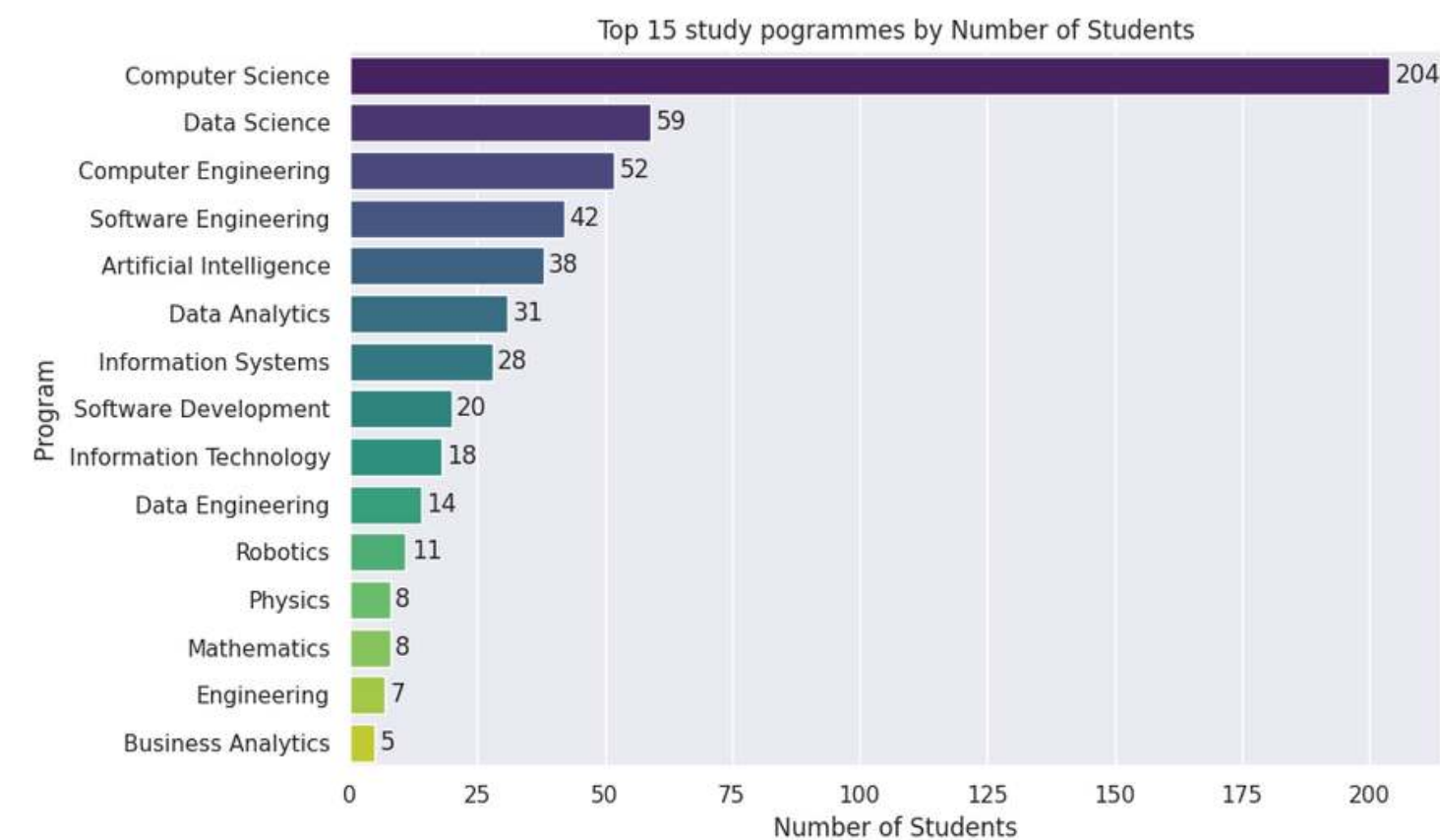


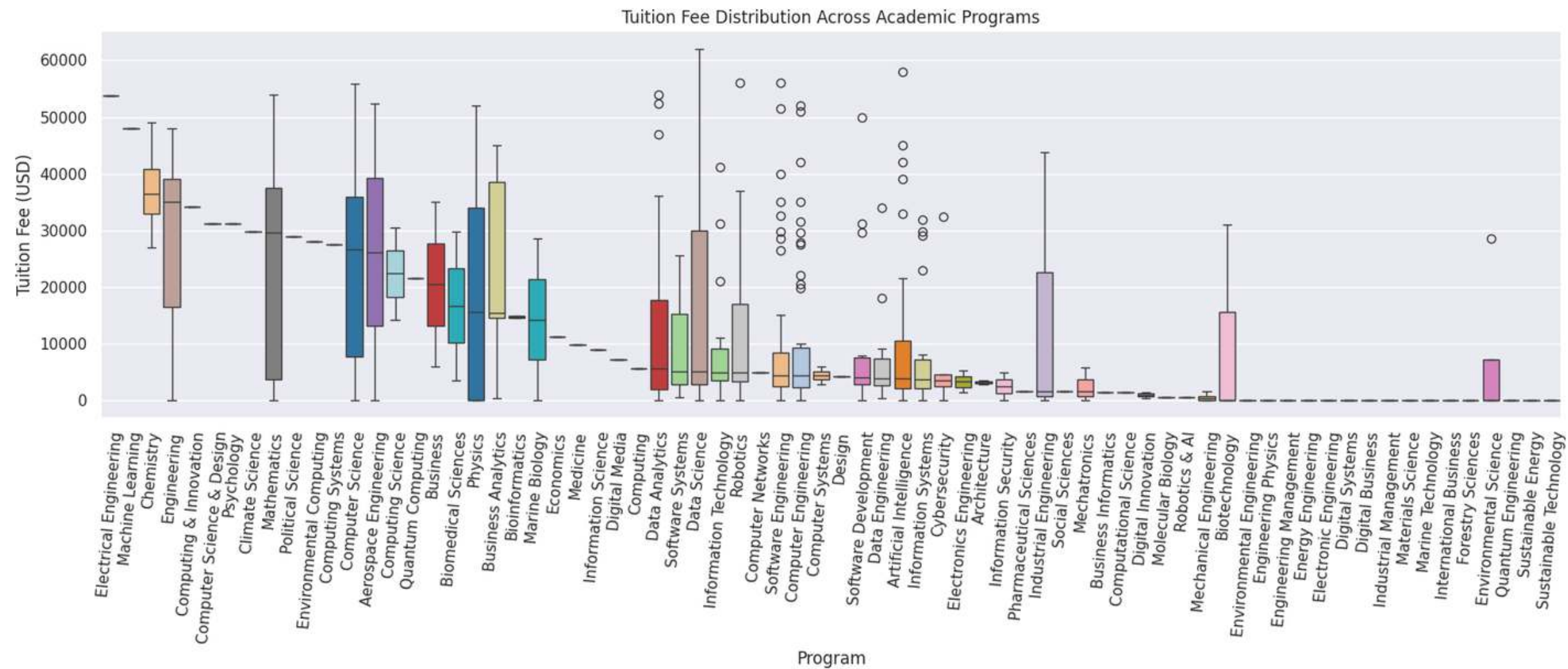
- **A Bachelor's degree has the highest median tuition** at roughly \$21,000.
- The median tuition for Master's (\$6,000) and PhD (\$5,500) programs is surprisingly low.
- When comparing three levels with the total cost, **Bachelor's remains the highest median total cost.**
- The data reveals that **tuition fees and total costs don't follow the same pattern across academic levels.**





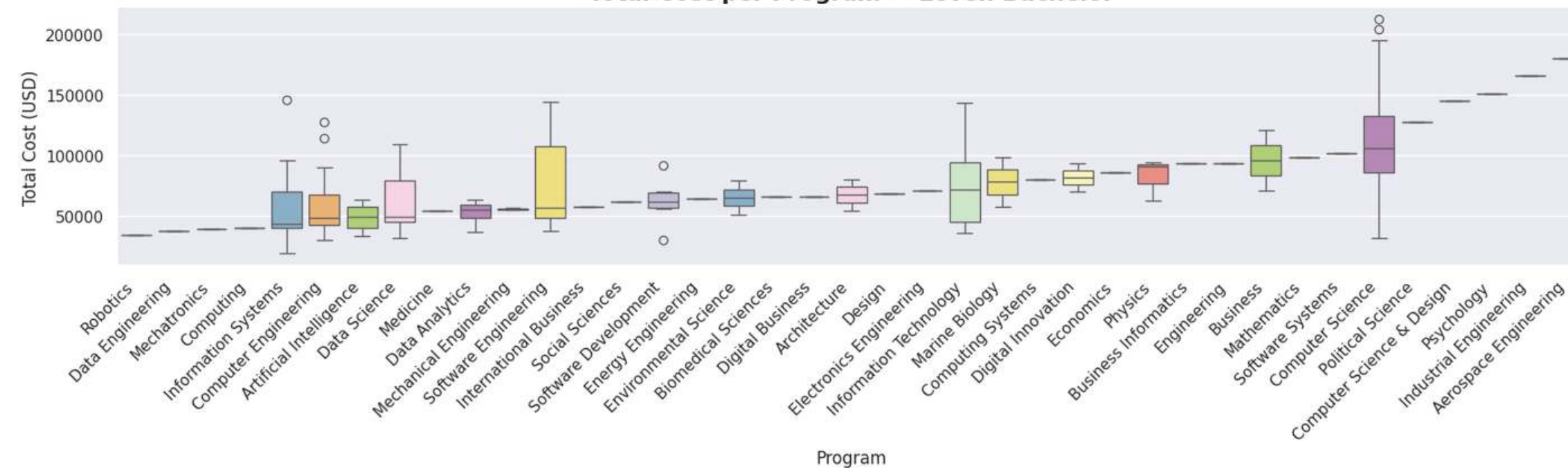
DEGREE PROGRAMS





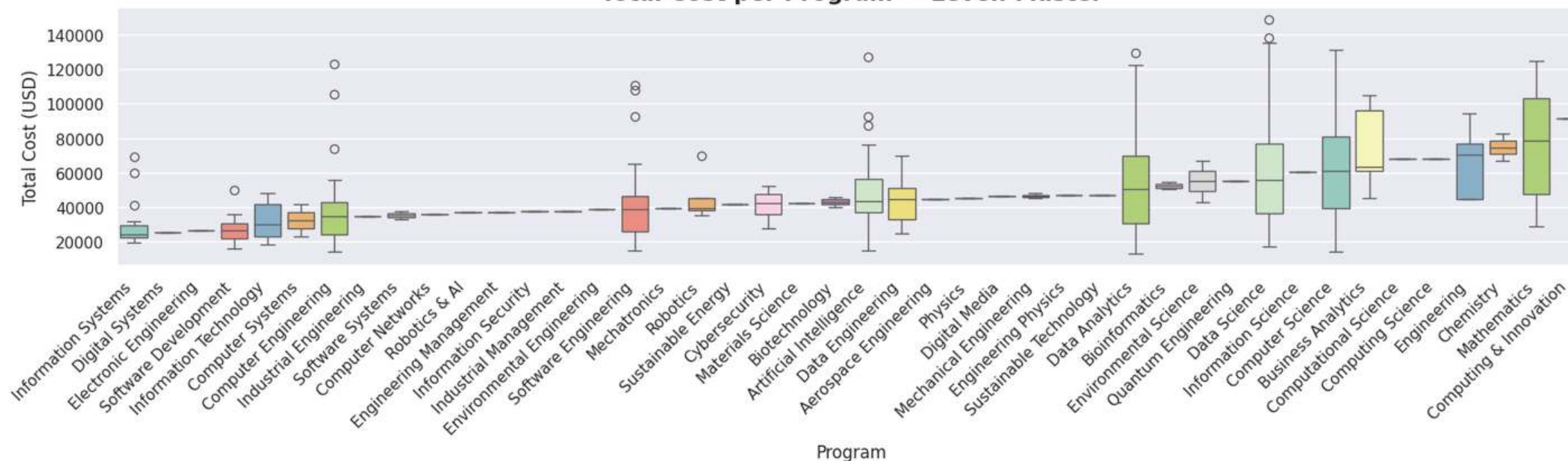
- Different academic programs have dramatically different tuition fee distributions
- Electrical Engineering, Machine Learning, and Chemistry consistently sit at the top of the price range, with medians appearing to be above \$40,000.
- Programs like Mathematics and Computer Science show massive vertical "whiskers."
- While some courses are affordable, others in the same field can cost upwards of \$50,000.(**Electrical Engineering VS Electronic Engineering**)
- Programs like Forestry Sciences, Marine Technology, and Sustainable Technology show almost no tuition cost.

Total Cost per Program — Level: Bachelor



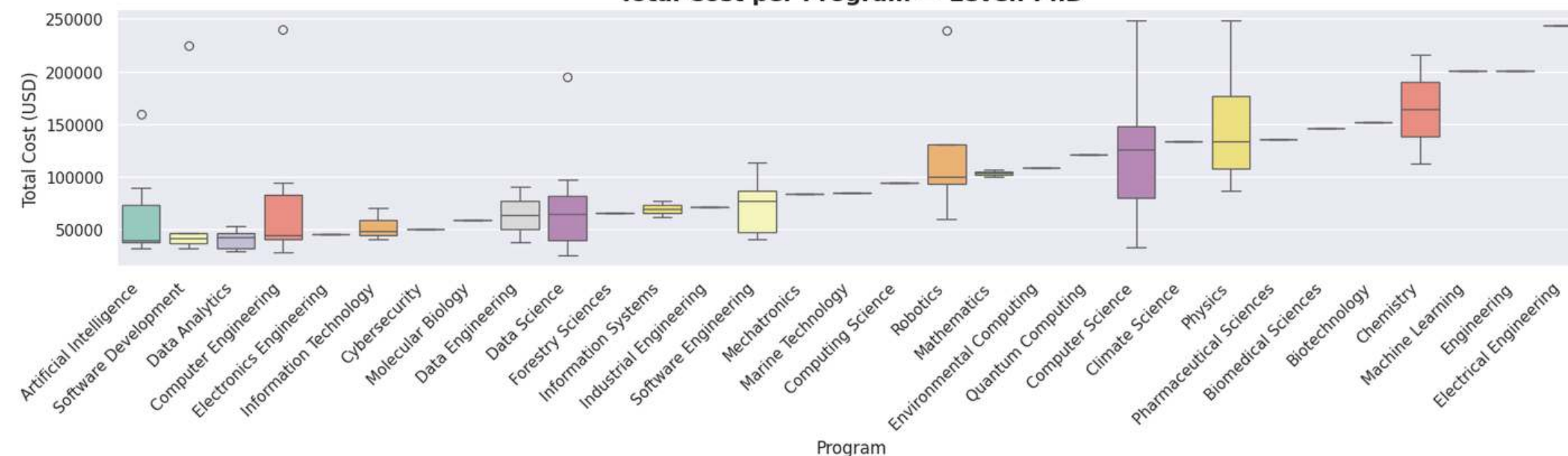
- **Bachelor Programs:** Most programs cluster between \$40,000 and \$100,000.
- Less variation compared to graduate programs

Total Cost per Program — Level: Master



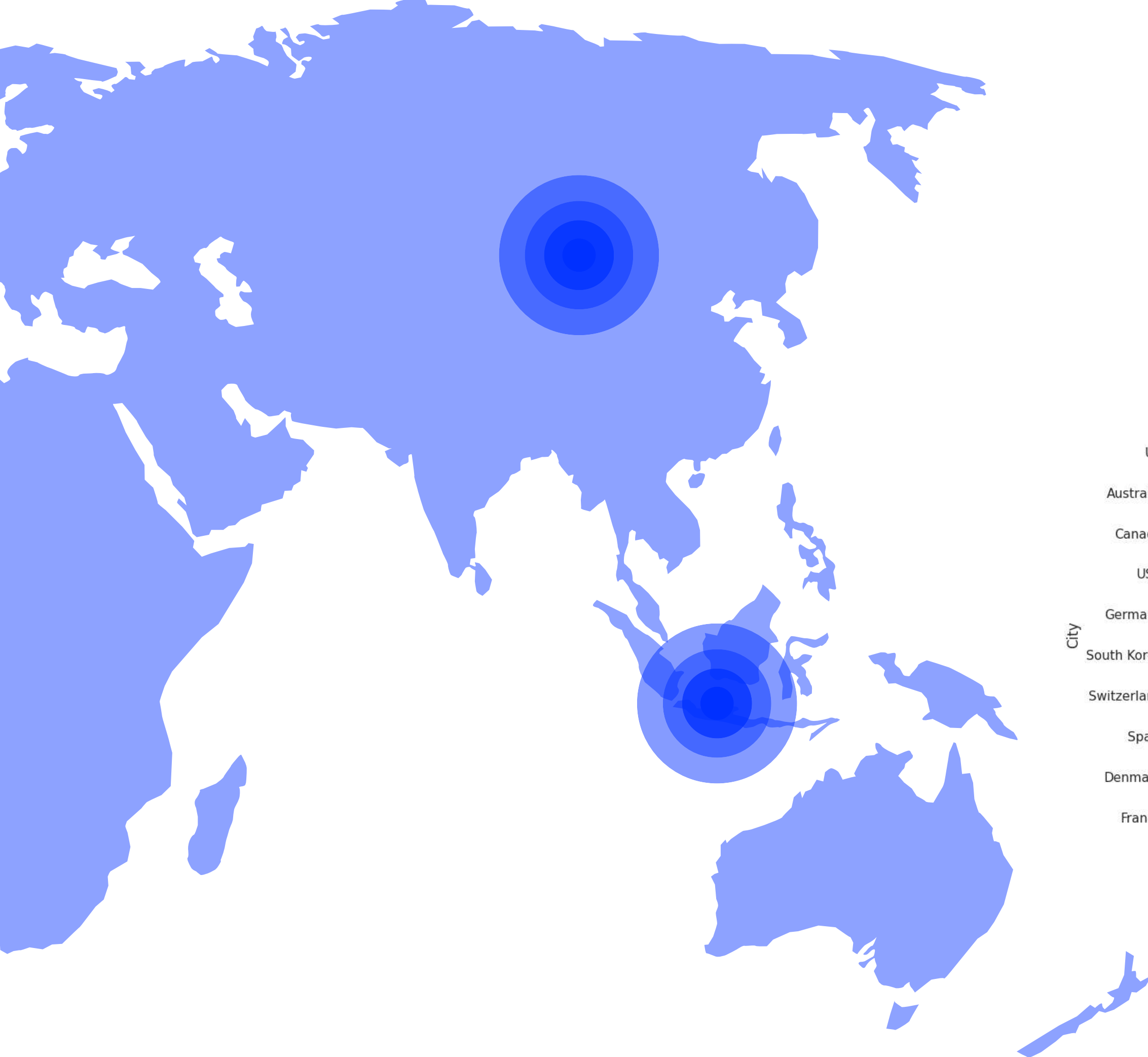
- **Master Programs:** These generally show a lower median cost than Bachelor programs, often ranging from \$25,000 to \$60,000. However, high-end outliers in Data Science and Information Science can push costs toward \$140,000.

Total Cost per Program — Level: PhD

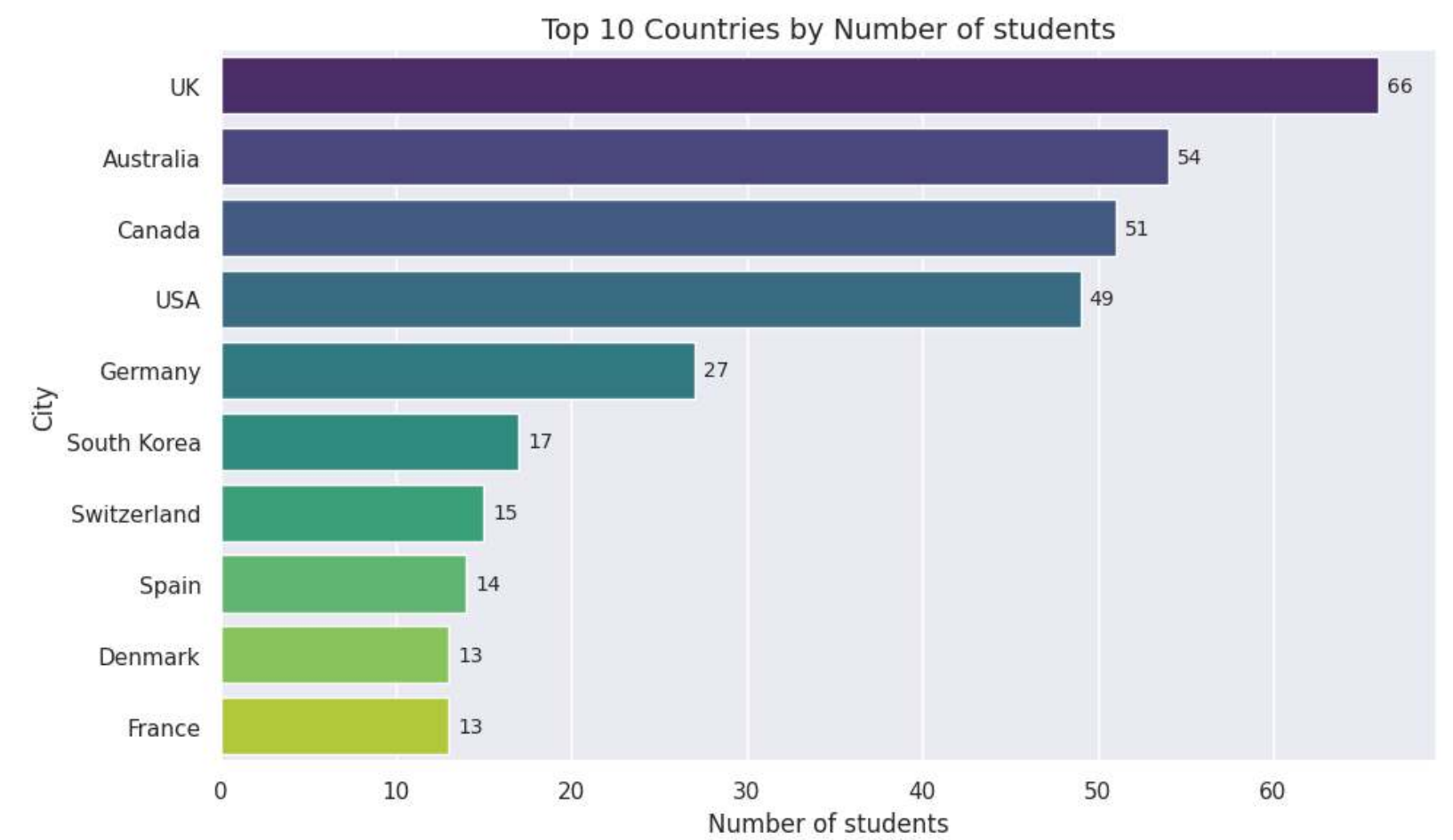


- **PhD Programs:** This level shows the highest volatility and some of the highest maximum costs. While many programs hover around \$50,000–\$100,000, specialized fields like Climate Science, Physics, and Chemistry show significant upward range, with some costs exceeding \$200,000.

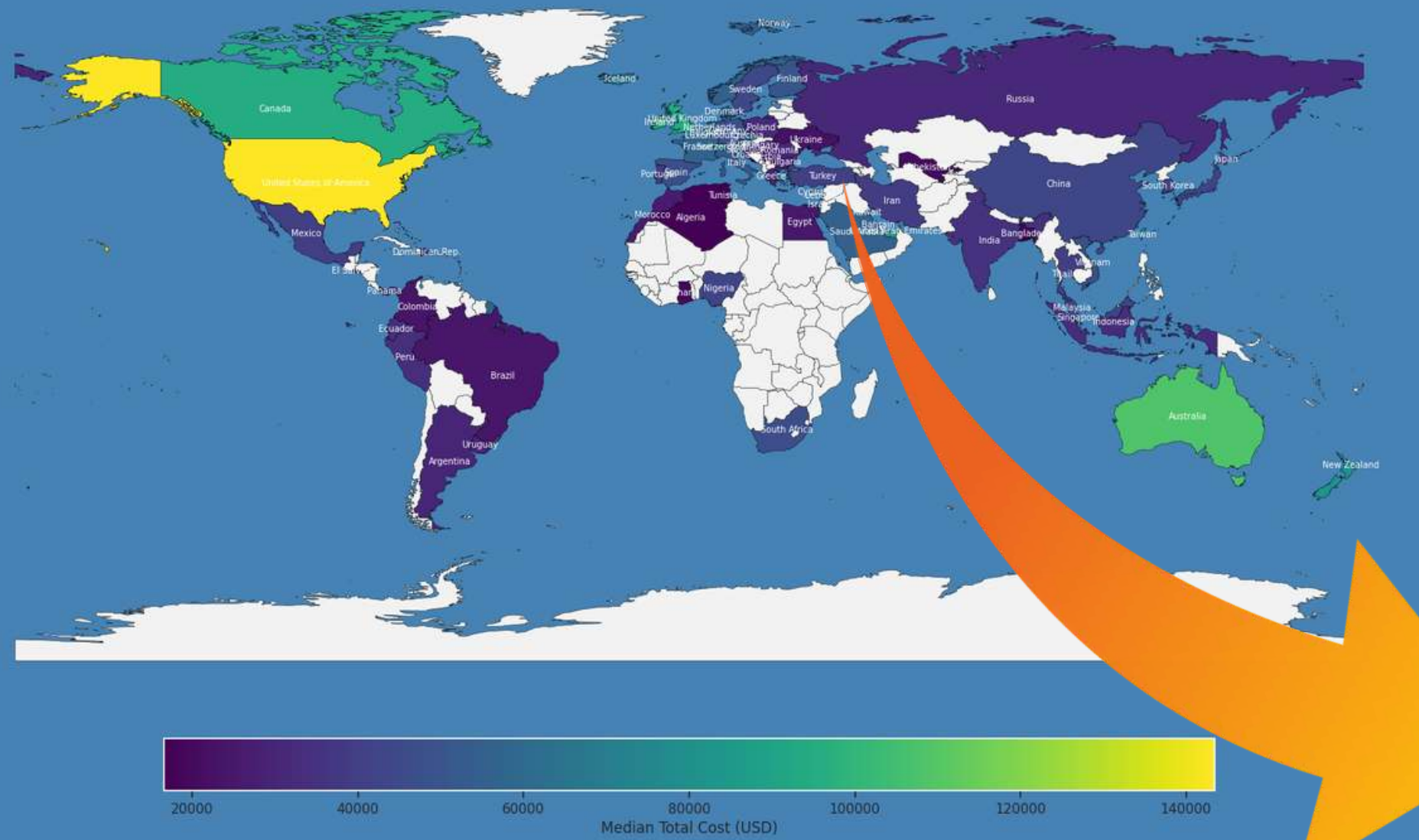
Even within the same degree level, there's enormous variation demonstrating that the field of study matters as much as the degree type.



COUNTRY



Global Distribution of Median Total Cost per Country



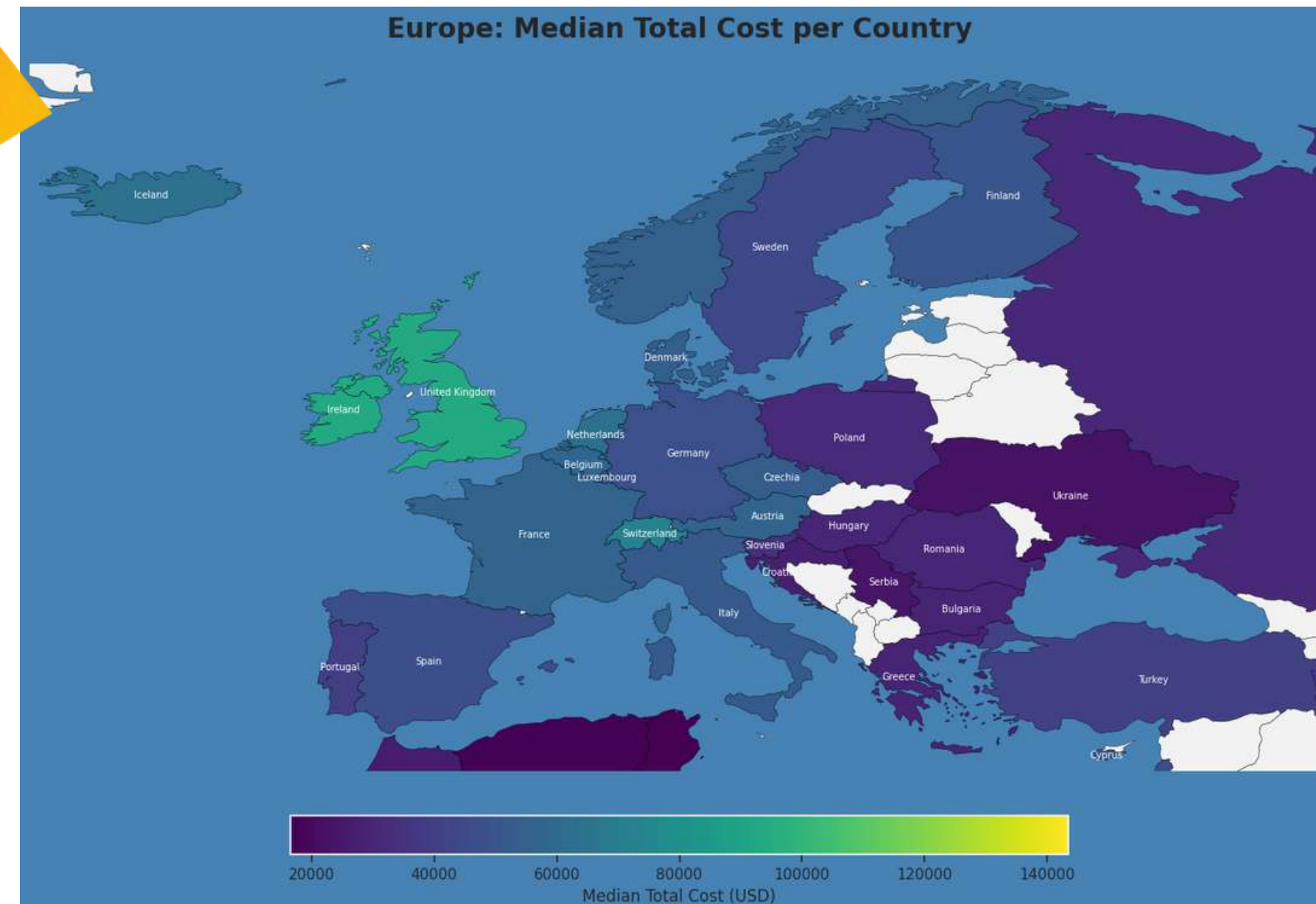
- **Highest Cost Regions:**

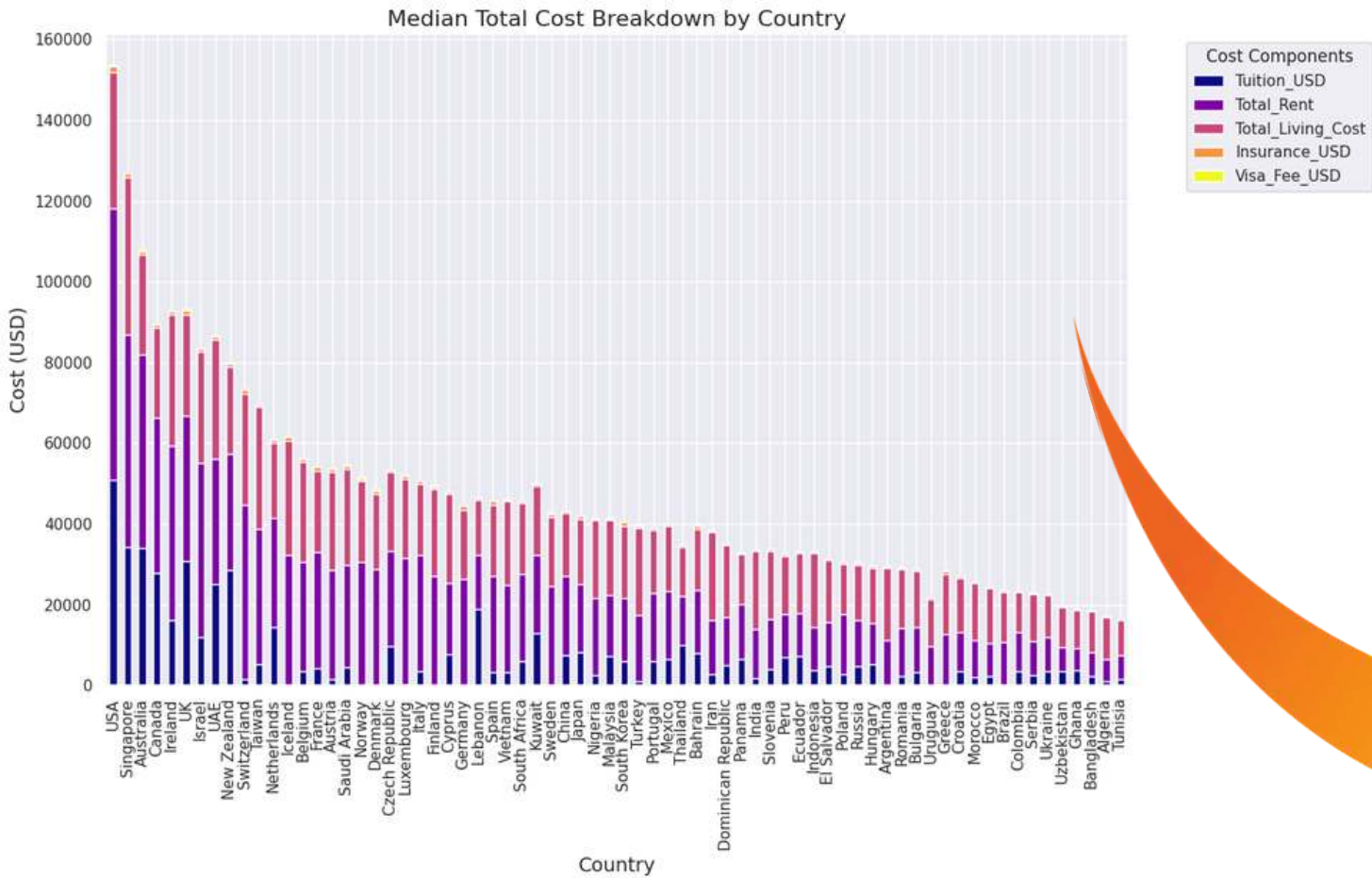
- The United States stands out as the most expensive region on the map, indicating a median total cost exceeding \$140,000.
- Countries like Canada and Australia are shaded in green/light green, suggesting costs in the \$80,000 to \$100,000 range.

- **Lower Cost Regions:**

- A significant portion of South America (e.g., Brazil, Argentina), Africa (e.g., Nigeria, Egypt), and Asia (e.g., India, China, Russia), as well as many countries in Europe, are shaded in dark purple or dark blue, suggesting moderate-to-low costs .

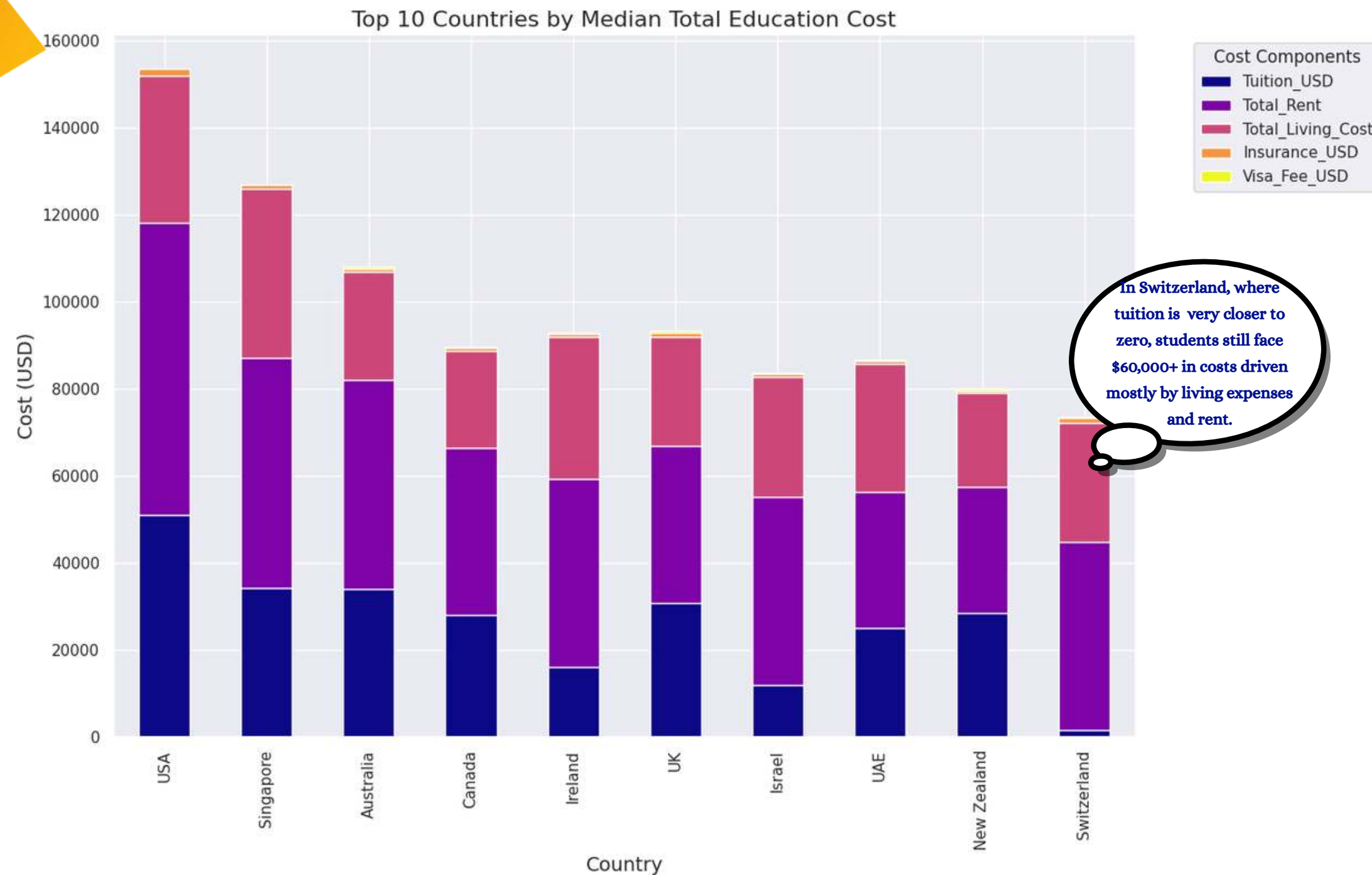
Europe: Median Total Cost per Country

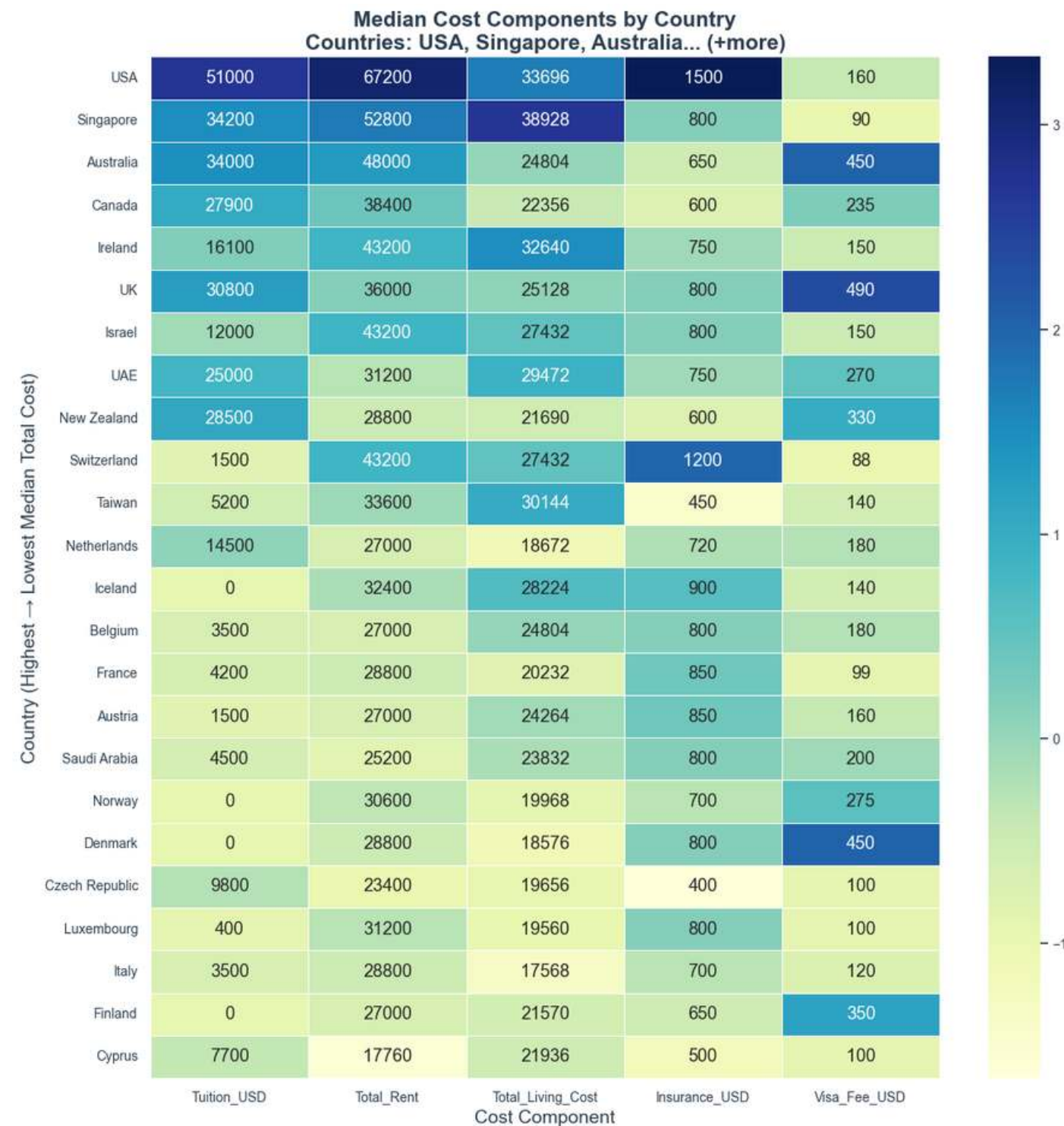




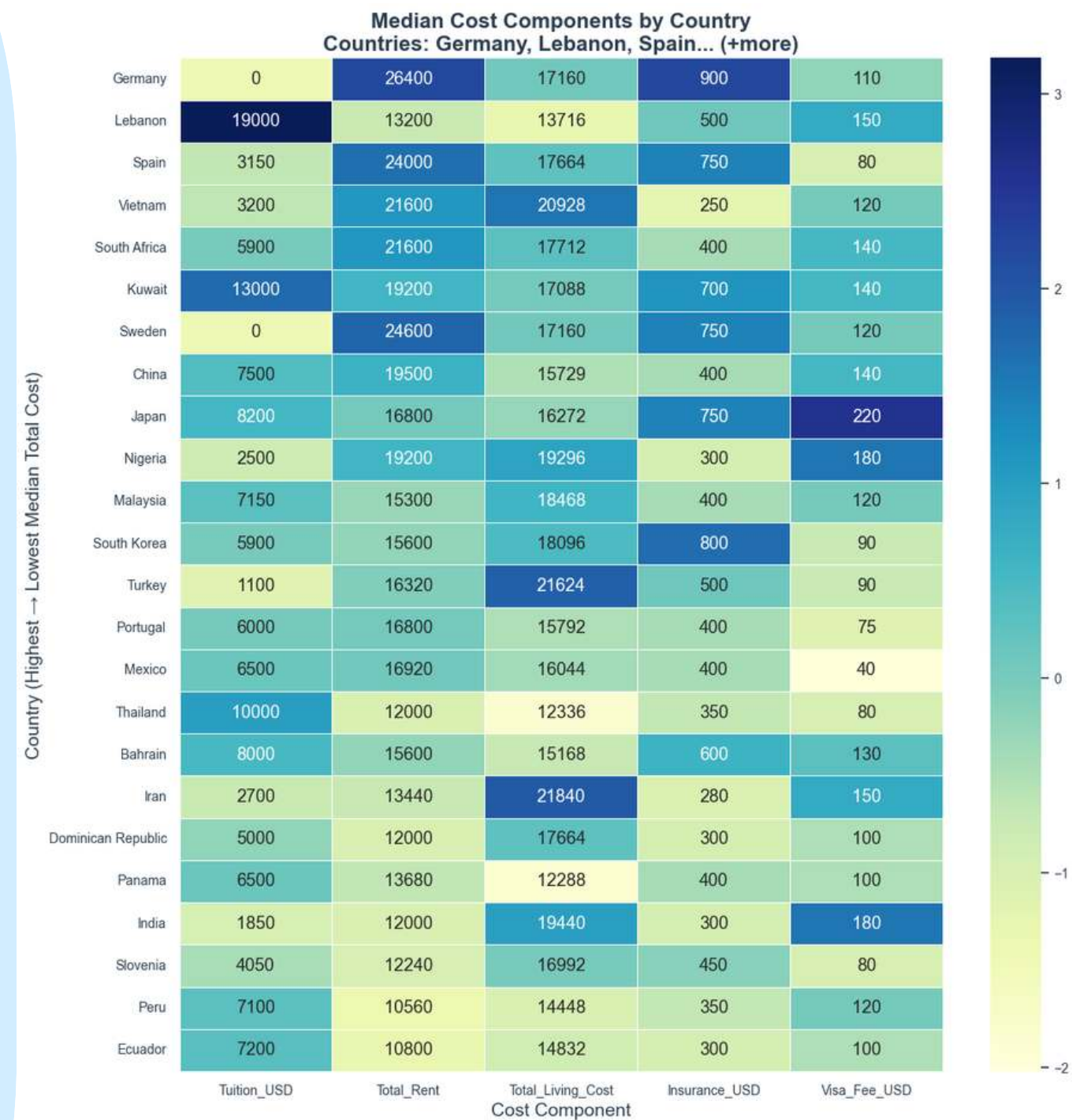
What are the primary cost drivers (the components contributing most to the total expenditure) for each countries ?

Compared to Insurance_USD, Tuition_USD, and Visa_Fee_USD, the two dominant cost drivers are clearly Rent_USD and Living_cost_USD, which make up the majority of total education costs across all countries.





- **Total_Living_Cost column** → consistently dark blue across most countries → **major driver**
- **Rent_USD column** → significant darkness → **another major driver**
- **Tuition_USD varies:** dark for USA/Singapore, pale/zero for Switzerland/Norway
- **Insurance & Visa_Fee** → mostly pale/narrow → **minimal impact**



To clearly see how the five cost drivers contribute to the response

“Total_Cost_USD” we can also use these heat maps as well.

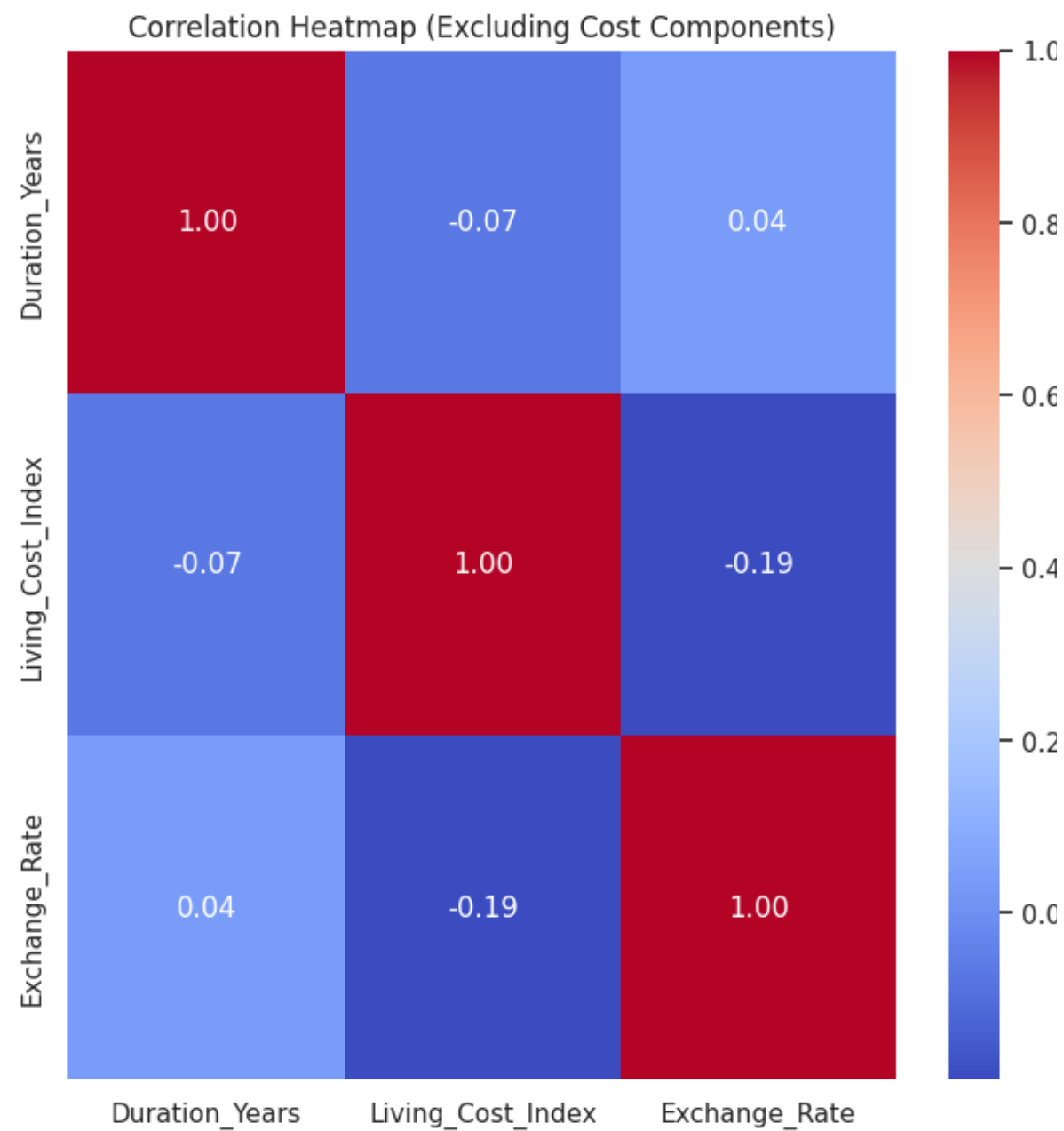
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EDA SUMMARY

- PhD shows the highest total cost despite lower tuition → duration & living impact.
- Total cost varies widely by field; engineering & tech programs are consistently high.
- USA is the most expensive; Canada & Australia high; many Asian, African & European countries are lower cost.
- Total cost depends on Country, Program, and Academic Level, supporting a multivariable prediction model.

”



MULTICOLLINEARITY

No Multicollinearity exists

	Feature	VIF	
0	Duration_Years	6.585970	} < 10
1	Living_Cost_Index	6.491912	
2	Exchange_Rate	1.037384	

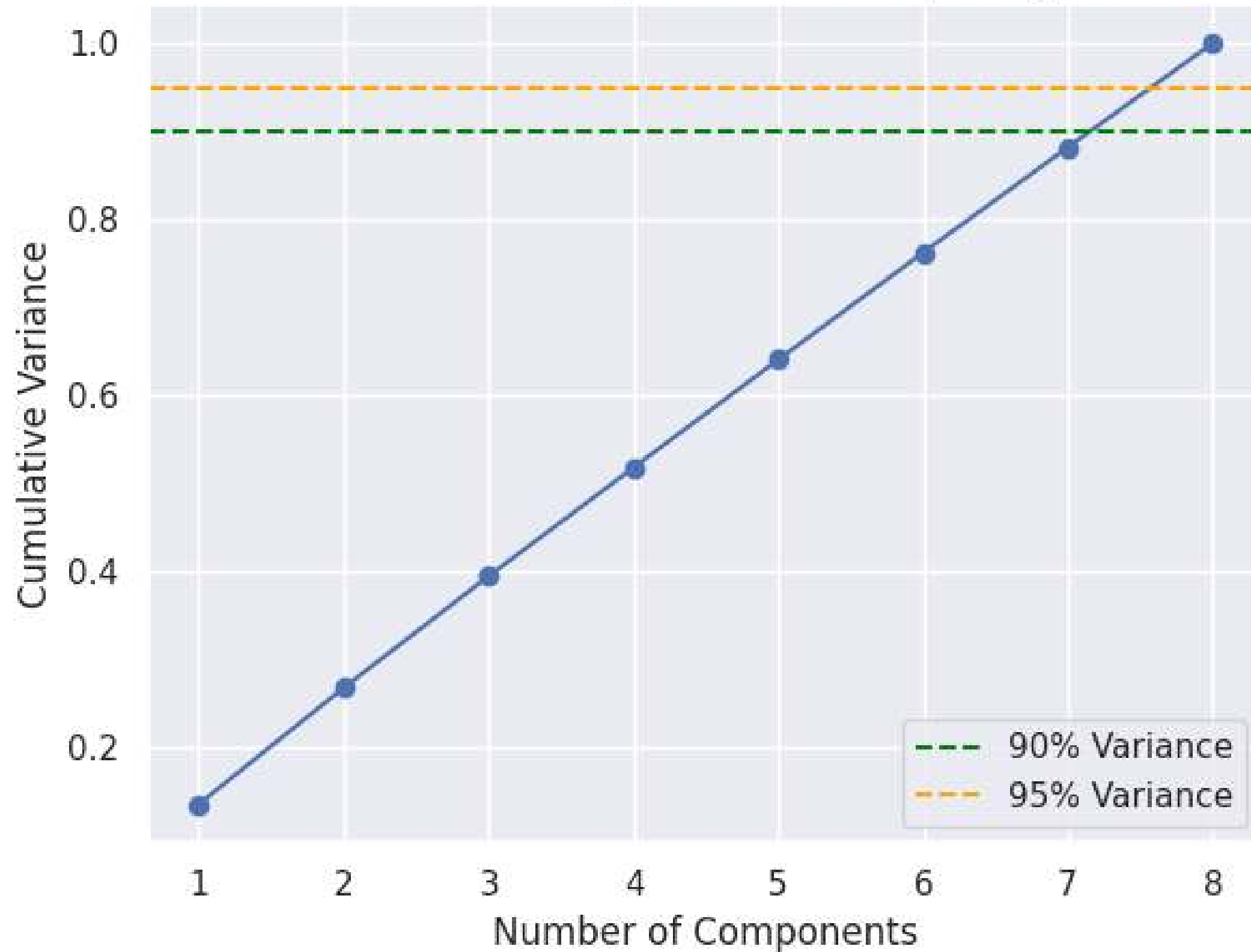
	Variable	Number_of_Outliers	Percentage_of_Outliers (%)
0	Duration_Years	0	0.00
1	Tuition_USD	0	0.00
2	Living_Cost_Index	20	2.21
3	Rent_USD	3	0.33
4	Visa_Fee_USD	93	10.25
5	Insurance_USD	78	8.60
6	Exchange_Rate	169	18.63
7	Total_Cost_USD	18	1.98

OUTLIER HANDLING

The IQR (Interquartile Range) method was used to count the number of outliers. During the modeling we phase we plan on dealing with the outliers by **removing the ones with lower percentage** and **using a suitable transformation for the others**

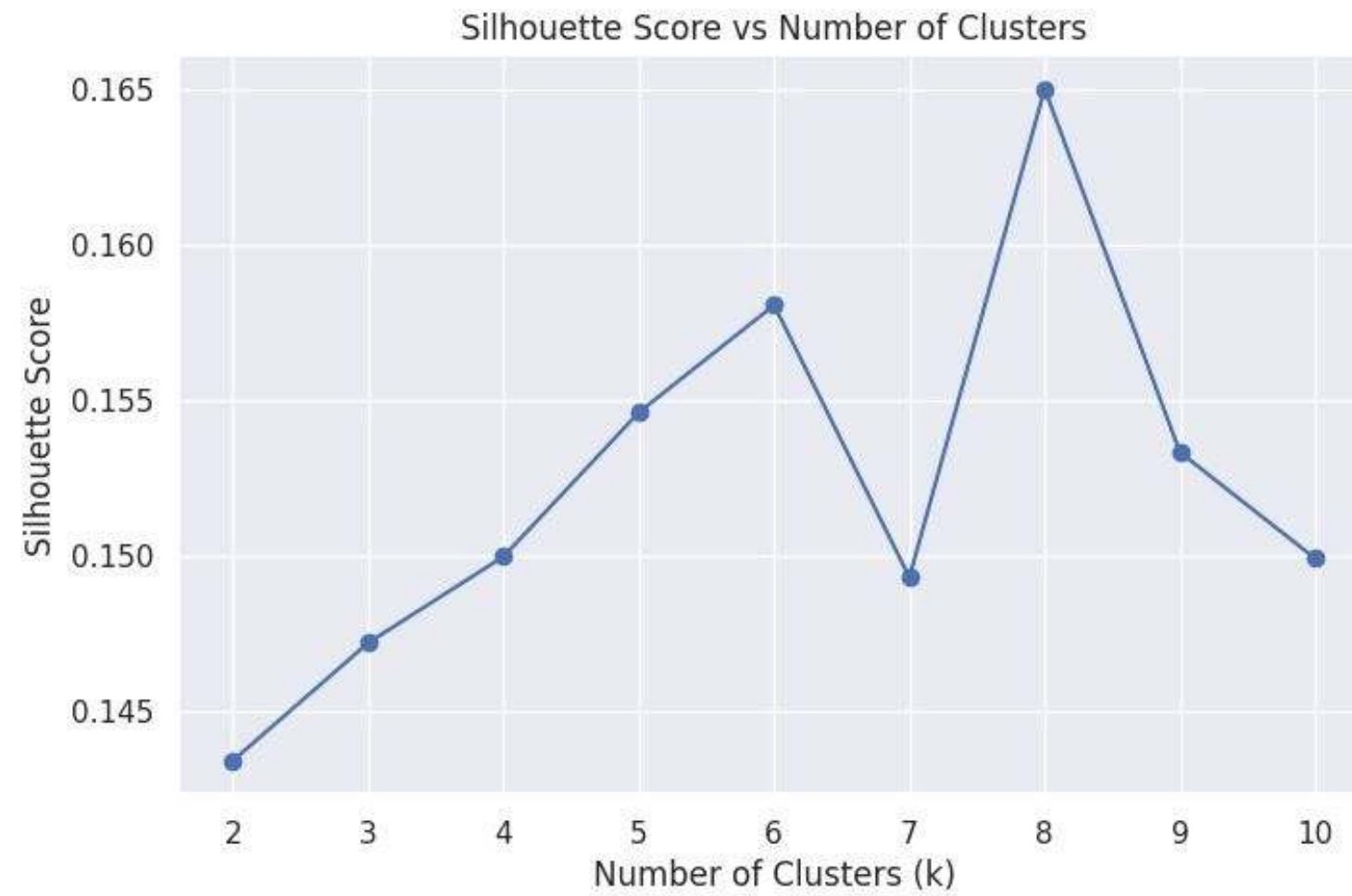
FAMD[•]

Cumulative Explained Variance (FAMD)



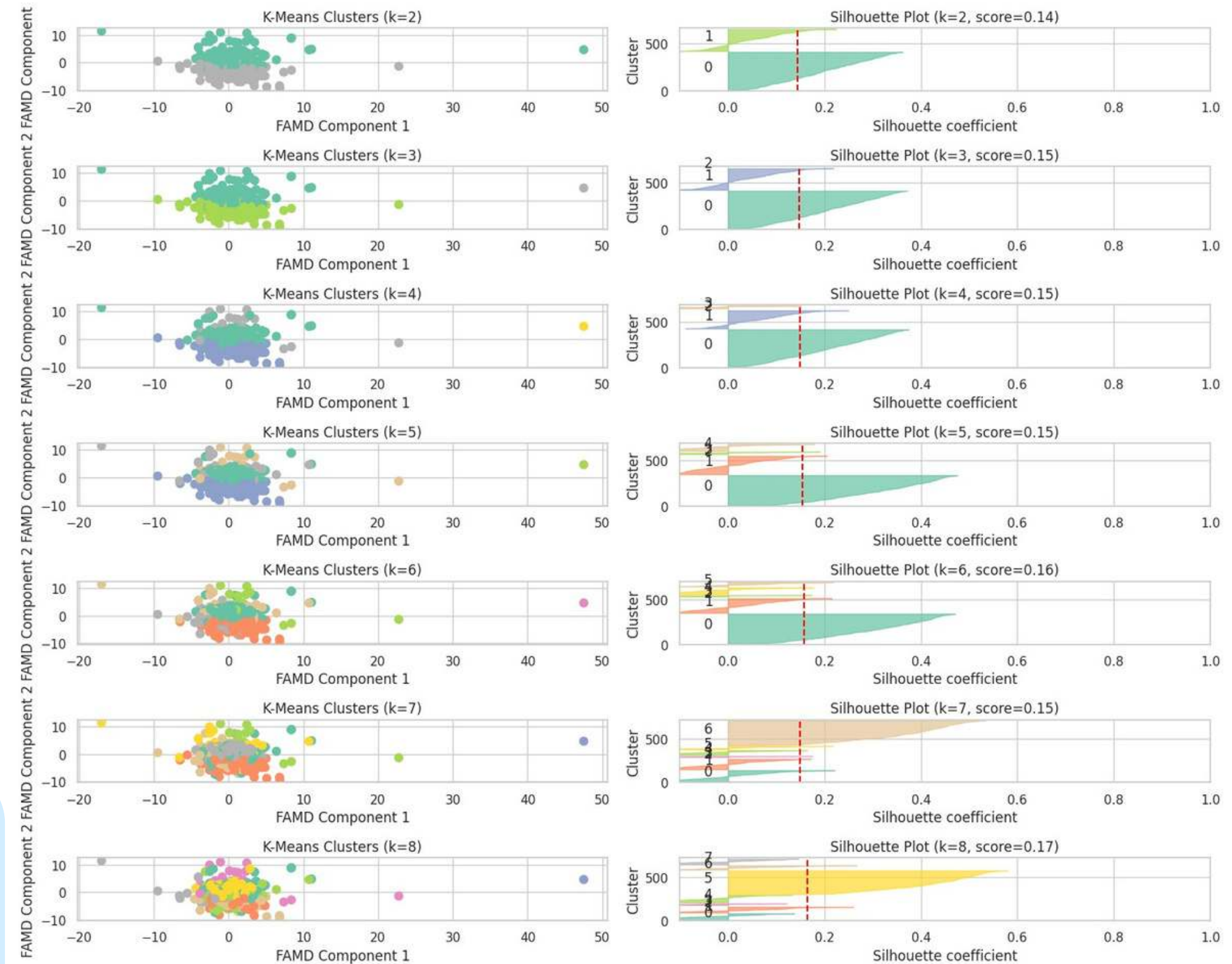
Looking at the cumulative explained variance, it shows that to explain 90% variance, at least 7 factors are required

So moving forward with factors would be pointless as there is no significant reduction in dimensions



CLUSTERS ●

A silhouette score greater than 0.5 typically indicates well-separated clusters. In this analysis, we evaluated silhouette scores for K values ranging from 2 to 8 using the K-means clustering method. The results indicated that all scores were below 0.2, suggesting that the clusters overlap. Therefore, there are no significant clusters present in the dataset.



“



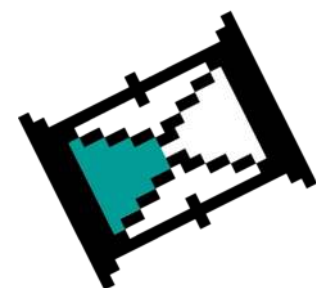
Plan for Advanced Analysis

- We plan to predict the total cost of education for a student.
- We will build and compare the performance of the following models: Linear Regression, Ridge, Lasso, Elastic Net, Decision Tree, Random Forest, and Boosted Trees, using stratified cross-validation.
- We will also try to perform multi-target prediction, so students can understand not only the total cost but also its individual cost components.

”



ONE MORE
THING



• EDUCOST CARD BUILDER

Proposed Data Product

An interactive tool to customize ready made components to enable to widen reach of awareness of education costs for international students.

This is a proposed product and features may be updated after further analysis and validation.

 **EduCost Card Builder**
Build education cost cards for your website

Live Preview

This is how the card will appear on your website

Technical University of Munich

Master's

Master of Computer Science

Munich Germany

Duration

2 years

Tuition

\$3,000

Monthly Rent

\$850

Visa Fee

\$75

Insurance

\$1,200

Estimated Total Cost

\$25,875

Including tuition, rent, visa & insurance for 2 years

The total cost is an estimate and may vary.

</> Generated Code

Copy to your Next.js project

```
"use client"

import * as React from "react"
import { MapPin, Building2, GraduationCap, BookOpen, Sparkles, Clock, DollarSign, Home, Plane, Shield,
```

Toggle Fields

Choose which fields to display on the card

LOCATION

Country

ISO country name

Germany

City

City location

Munich

University

Institution name

Technical University of Munich

PROGRAM

Program

Course/major name

Master of Computer Science

Level

Degree level

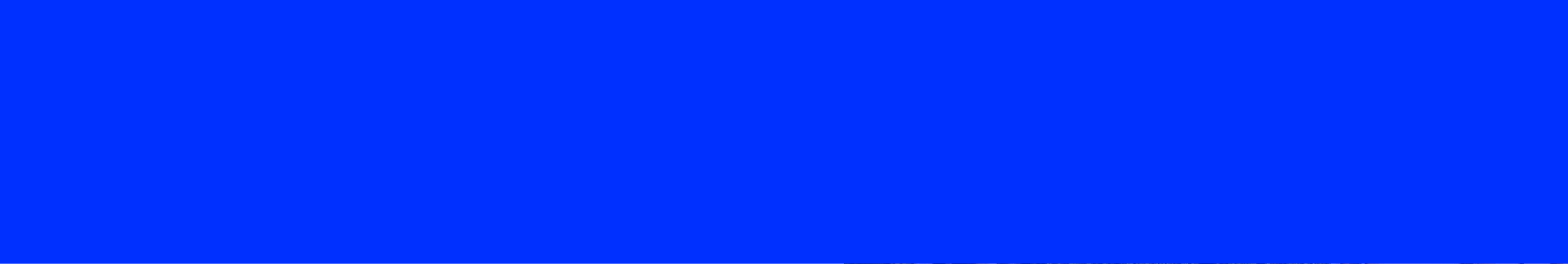
Master's

Duration

Program length in years

2

COST



THANK YOU

Stay tuned for the results of the
predictive modeling phase !

